DIODE CHARACTERISTIC CURVE

SEMICONDUCTOR DIODE

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Diode Approximations

Ideal, Second, and Third Approximation

Shockley's Equation

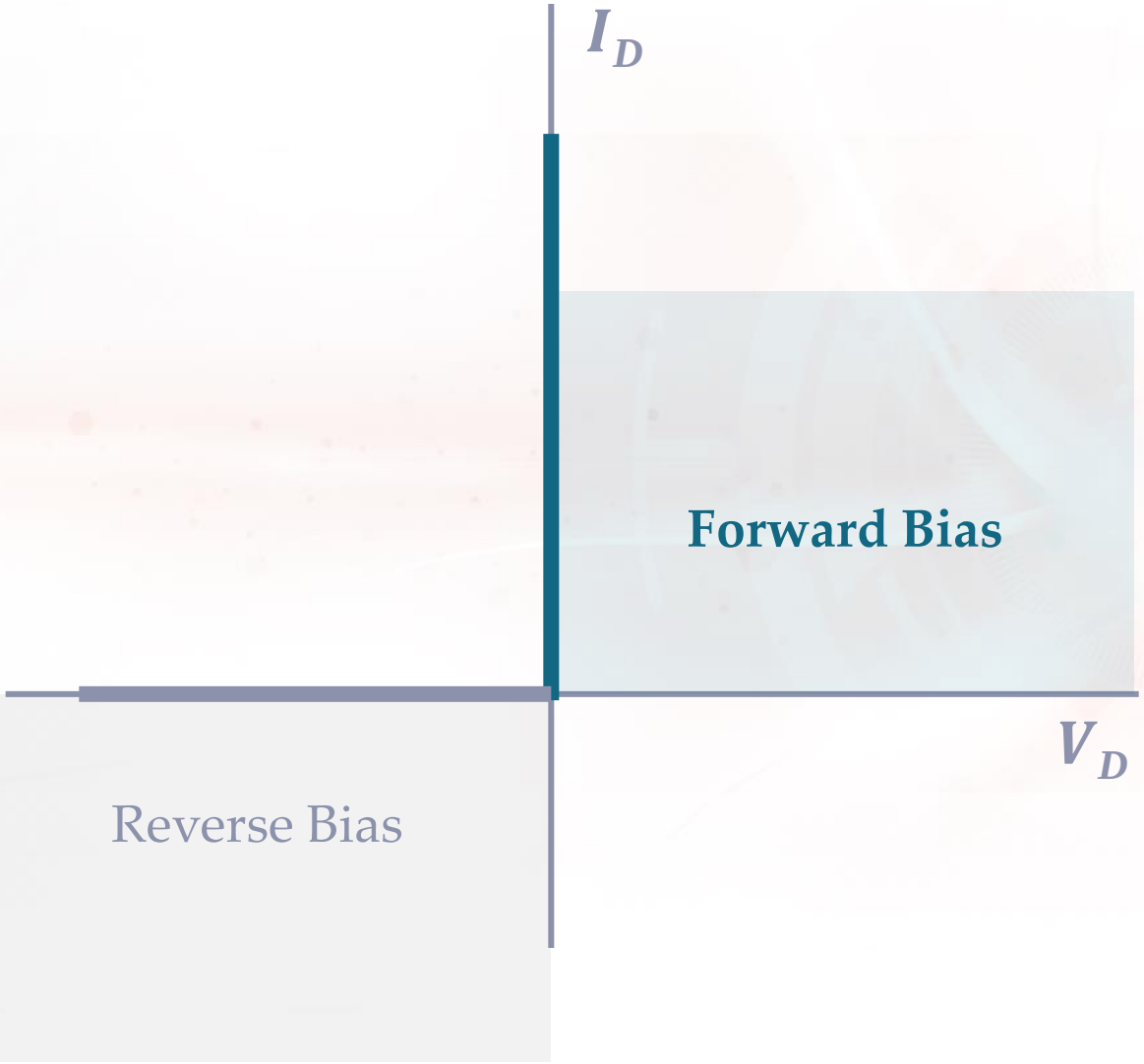
Load Line Analysis

Reading Data Sheet

DIODE APPROXIMATIONS

IDEAL DIODE

Characteristic Curve



Acts like a **switch**:

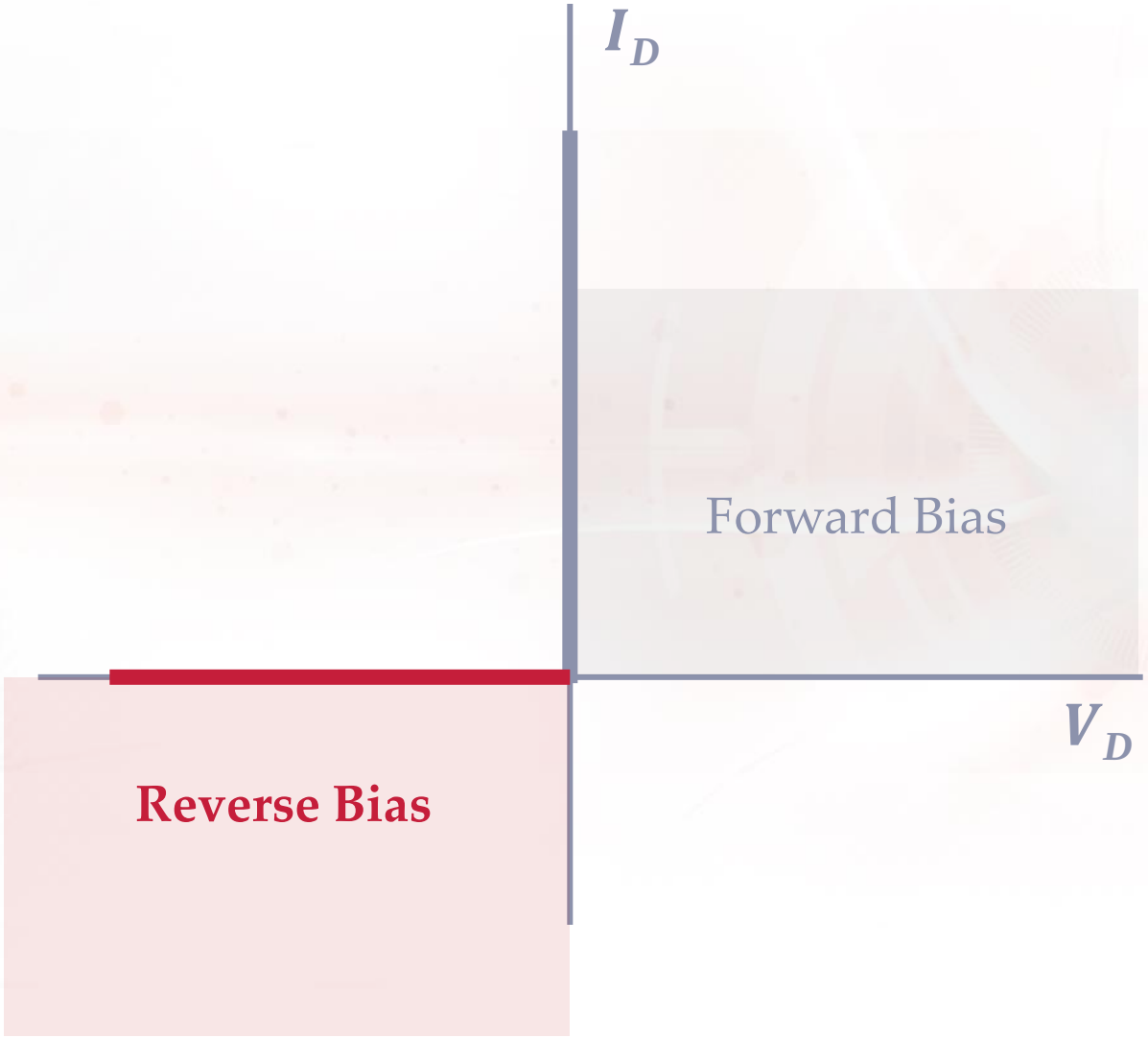
Zero (0) resistance when **forward** bias

Circuit Equivalent

Forward Bias

IDEAL DIODE

Characteristic Curve



Acts like a **switch**:

Infinite (∞)resistance when **reverse** bias

Circuit Equivalent

Reverse Bias

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.

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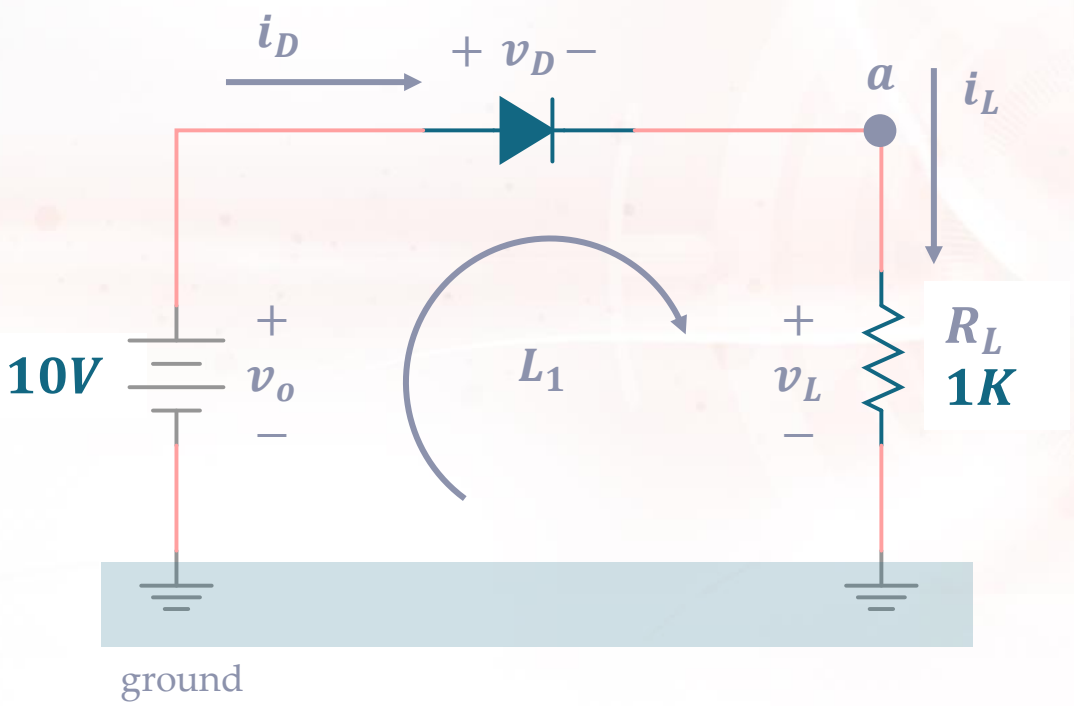


KVL @ L_1,

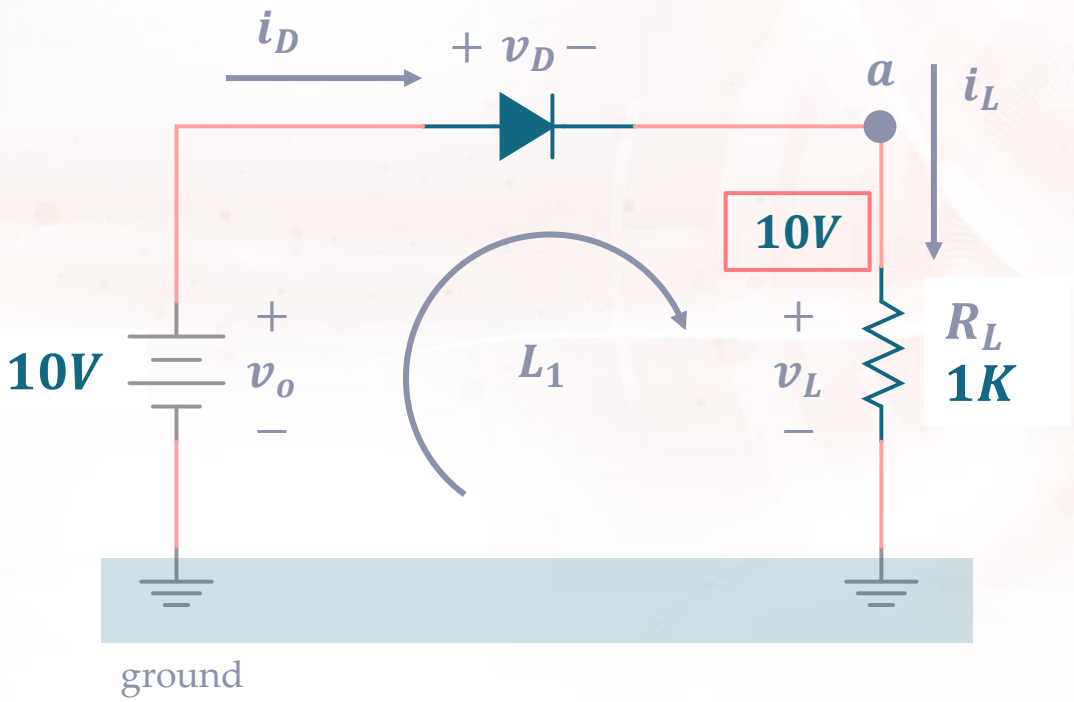
$$-v_0 + v_D + v_L = 0$$

$$v_L = v_o + \cancel{v_D}^{\,0}$$

$$\boxed{v_L = 10\,V}$$

ans

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.



KCL @ a,

$$-i_D + i_L = 0$$

$$i_D = i_L$$

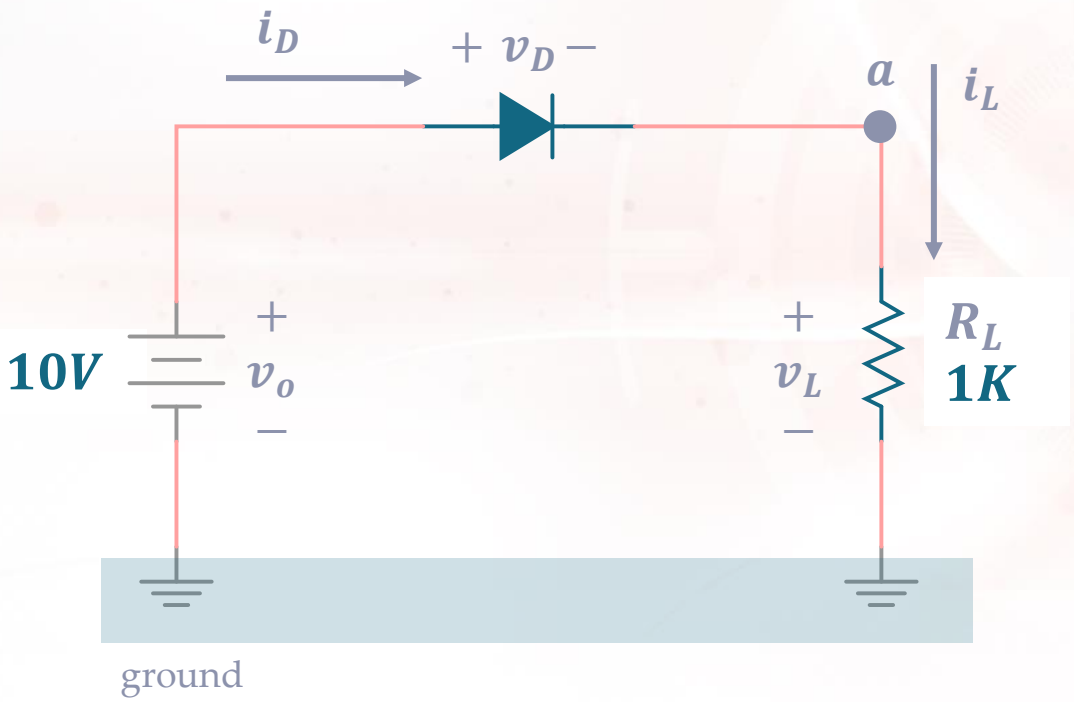
$$i_D = \frac{v_L}{R_L}$$

$$i_D = \frac{10}{1K}$$

$$\boxed{i_L = 10\ mA}$$

ans

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.



Node analysis method,

$$v_D = v_o - v_a$$

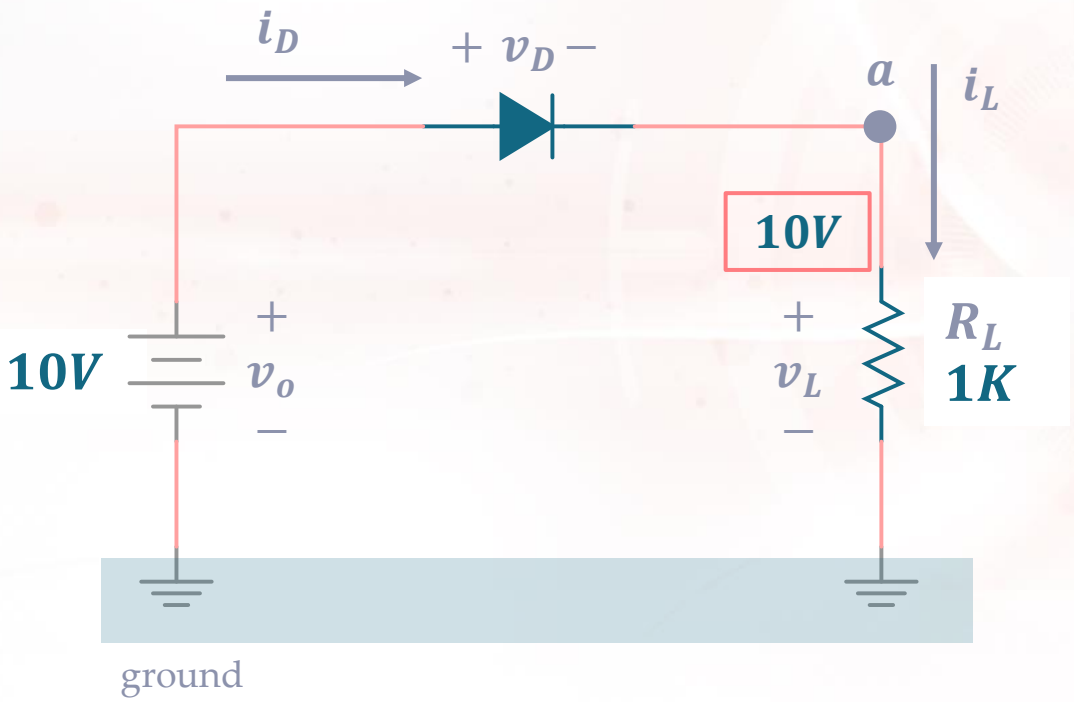
$$0 = 10 - v_a$$

$$v_a = 10$$

$$\boxed{v_L = 10\,V}$$

ans

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.



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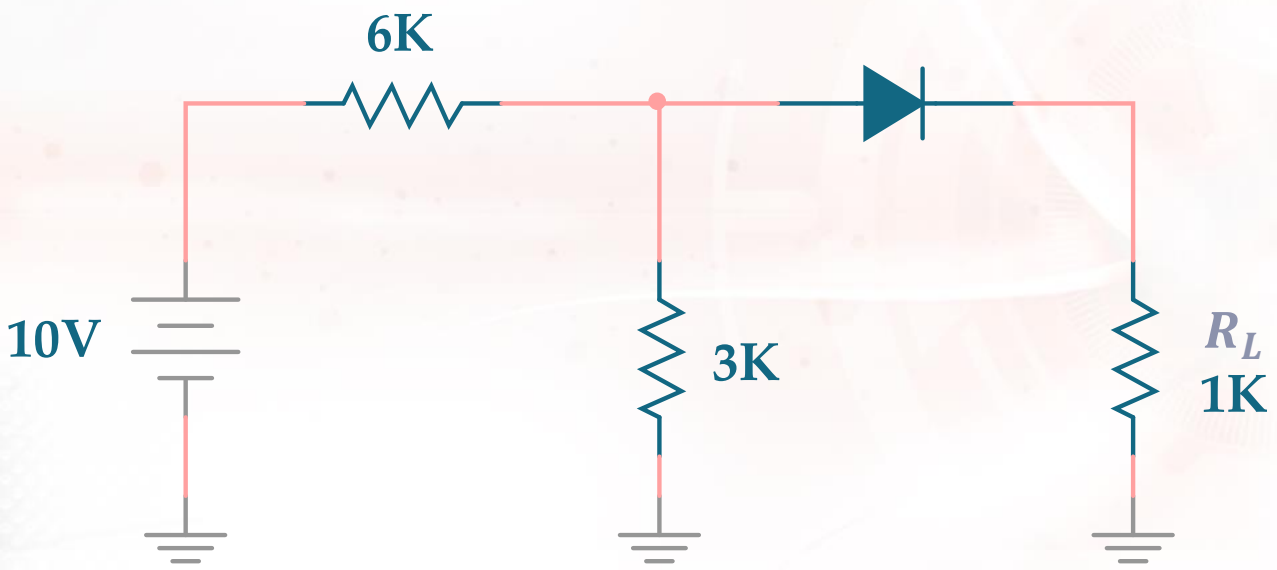

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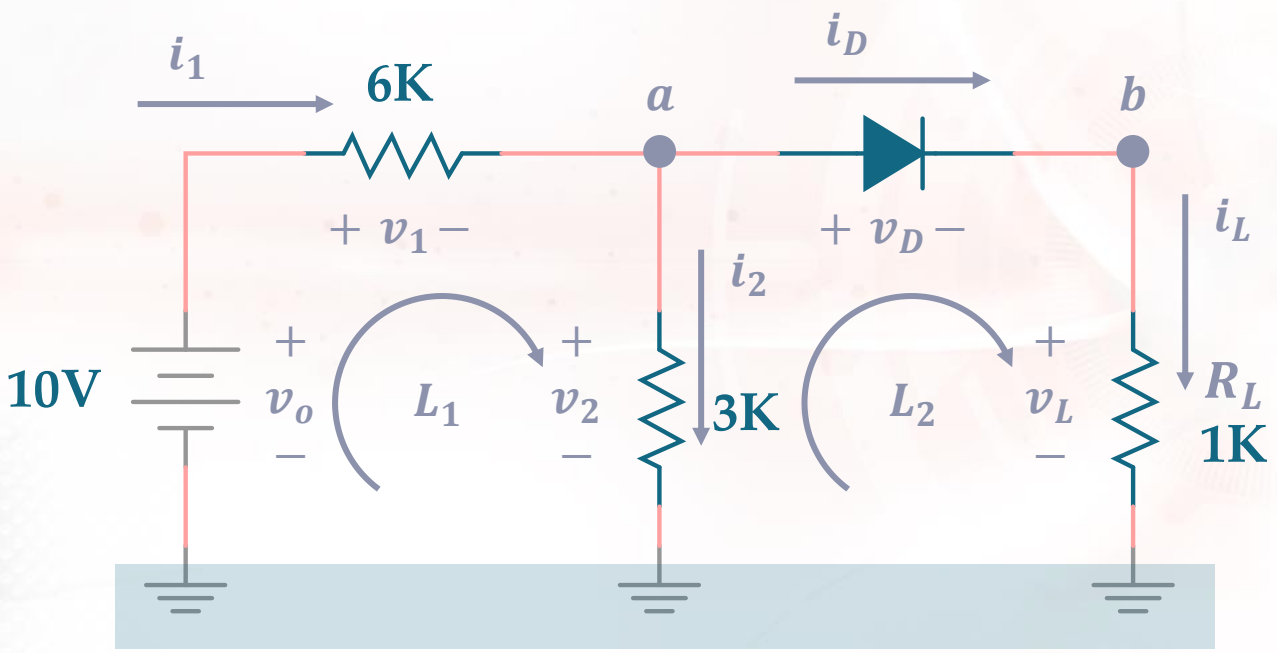


KCL @ a,

$$-i_1 + i_2 + i_D = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + i_D = 0$$

KCL @ b,

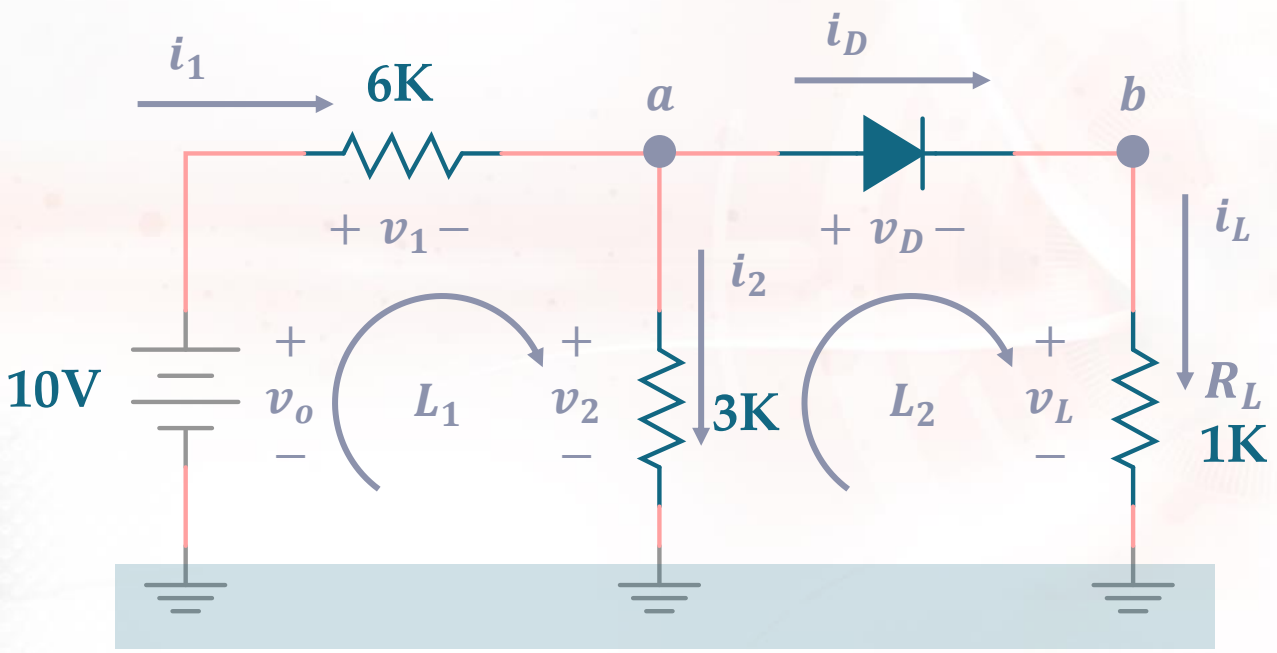
$$-i_D + i_L = 0$$

$$i_D = i_L$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + i_L = 0$$

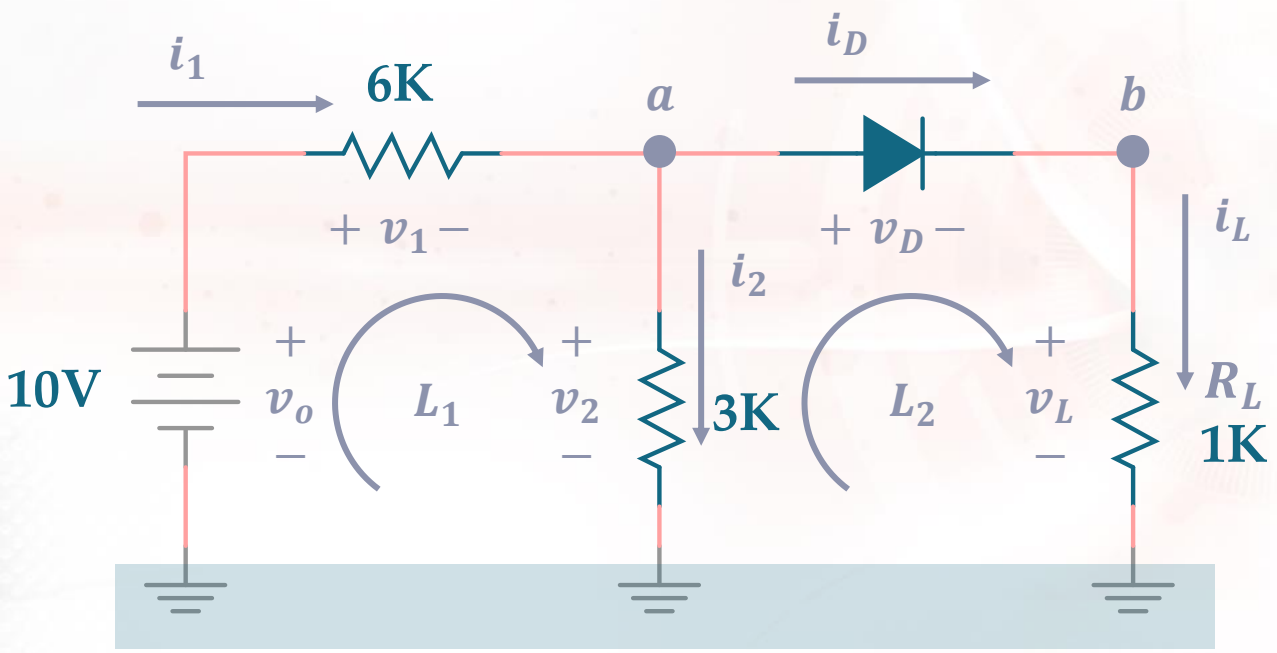
$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_L}{R_L} = 0$$

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.



$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_L}{R_L} = 0$$

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.



$$-\frac{v_1}{R_1}+\frac{v_2}{R_2}+\frac{v_L}{R_L}=0$$

1 $$-\frac{v_1}{6K}+\frac{v_2}{3K}+\frac{v_L}{1K}=0$$

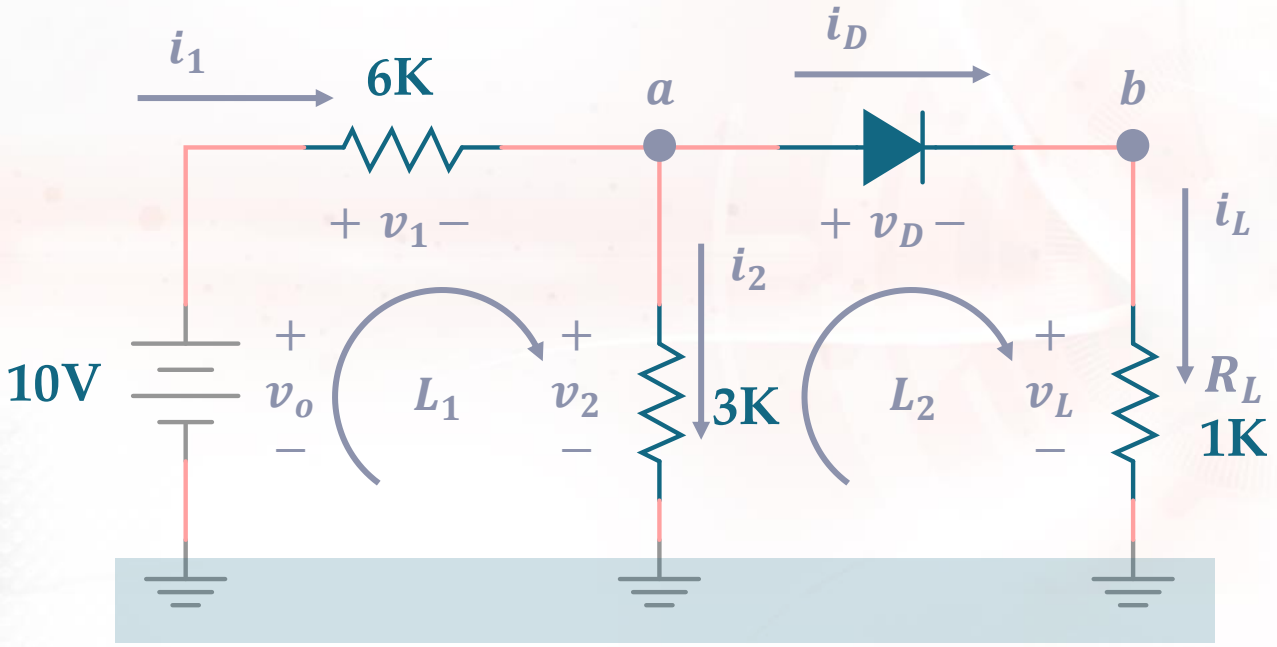
KVL @ L_1,

$$-v_o+v_1+v_2=0$$

$$-10+v_1+v_2=0$$

2 $$v_1+v_2=10$$

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.



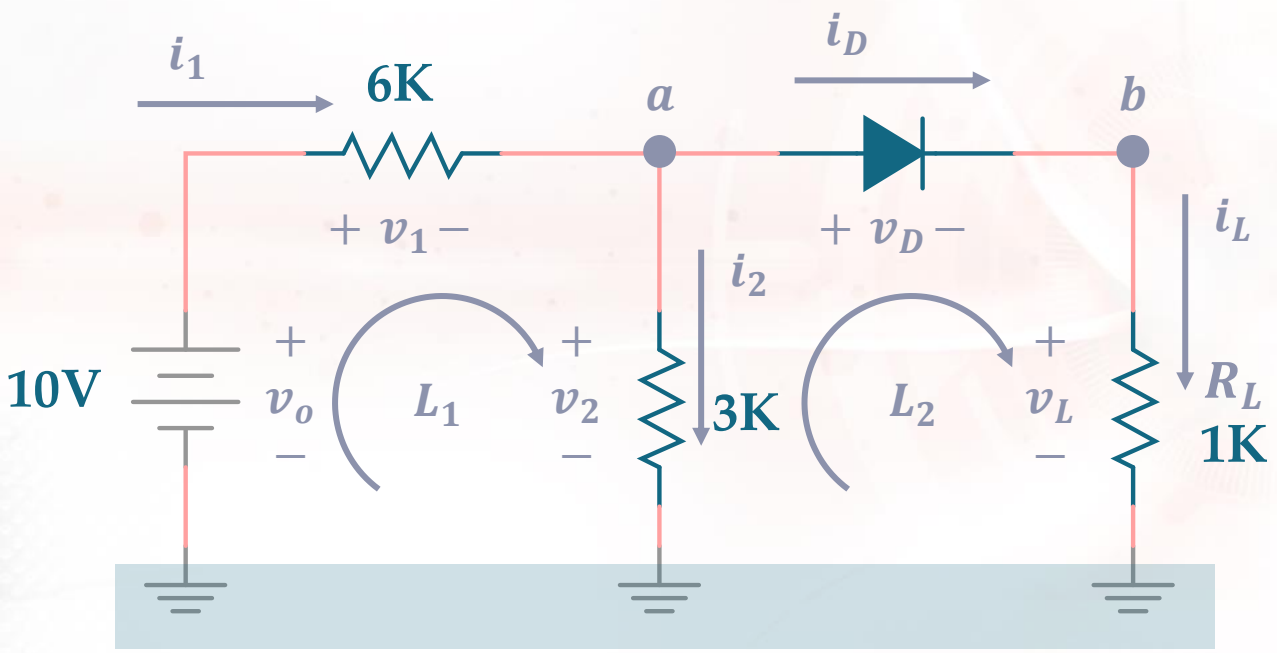
KVL @ L_2,

$$-v_2 + v_D + v_L = 0$$

$$-v_2 + 0 + v_L = 0$$

3 $$-v_2 + v_L = 0$$

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.



Linear Equations,

$$-\frac{v_1}{6K} + \frac{v_2}{3K} + \frac{v_L}{1K} = 0$$

$$v_1 + v_2 = 10$$

$$-v_2 + v_L = 0$$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 1 & 1 & 0 & 10 \\ 0 & -1 & 1 & 0 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

Linear Equations,

$$-\frac{v_1}{6K} + \frac{v_2}{3K} + \frac{v_L}{1K} = 0$$

$$v_1 + v_2 = 10$$

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Augmented Matrix,

$$
\begin{array}{cc}
\begin{array}{cccc} v_1 & v_2 & v_L & \end{array} & \\
\begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 1 & 1 & 0 & 10 \\ 0 & -1 & 1 & 0 \end{bmatrix} & \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}
\end{array}
$$

Augmented Matrix,

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$$\begin{array}{cc} \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} & \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 0 & 3 & 6 & 10 \\ 0 & -1 & 1 & 0 \end{bmatrix} & \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

$r_2 \rightarrow 6Kr_1 + r_2$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 0 & 3 & 6 & 10 \\ 0 & 0 & 9 & 10 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 0 & 3 & 6 & 10 \\ 0 & -1 & 1 & 0 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$r_3 \rightarrow 3r_3 + r_2$$

Augmented Matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} \\ & \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 0 & 3 & 6 & 10 \\ 0 & 0 & 9 & 10 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

from equation 3,

$$9v_L = 10$$

$$v_L = \frac{10}{9}$$

$$\boxed{v_L = 1.11\,V}$$

ans

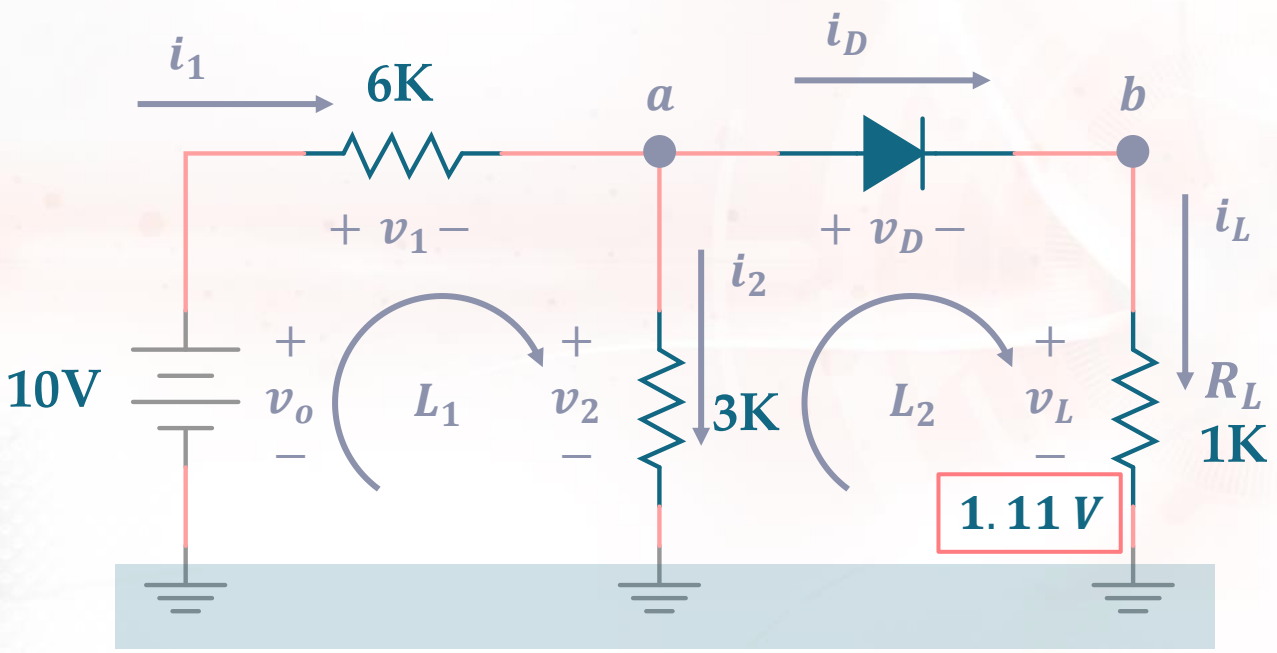
Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.

$$i_L = \frac{v_L}{R_L}$$

$$i_L = \frac{1.11}{1K}$$

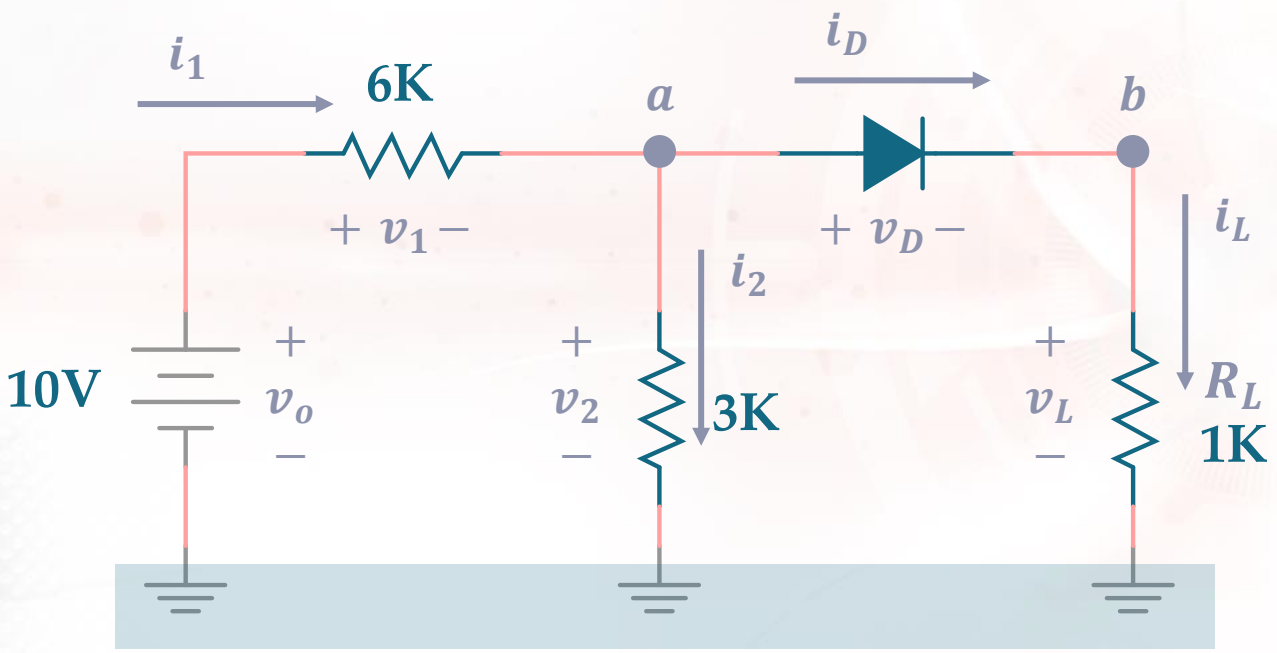
$$\boxed{i_L = 1.11\ mA}$$

ans



ground

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.



Node Analysis Method,

KCL @ a,

$$-i_1 + i_2 + i_D = 0$$

$$-i_1 + i_2 + i_L = 0$$

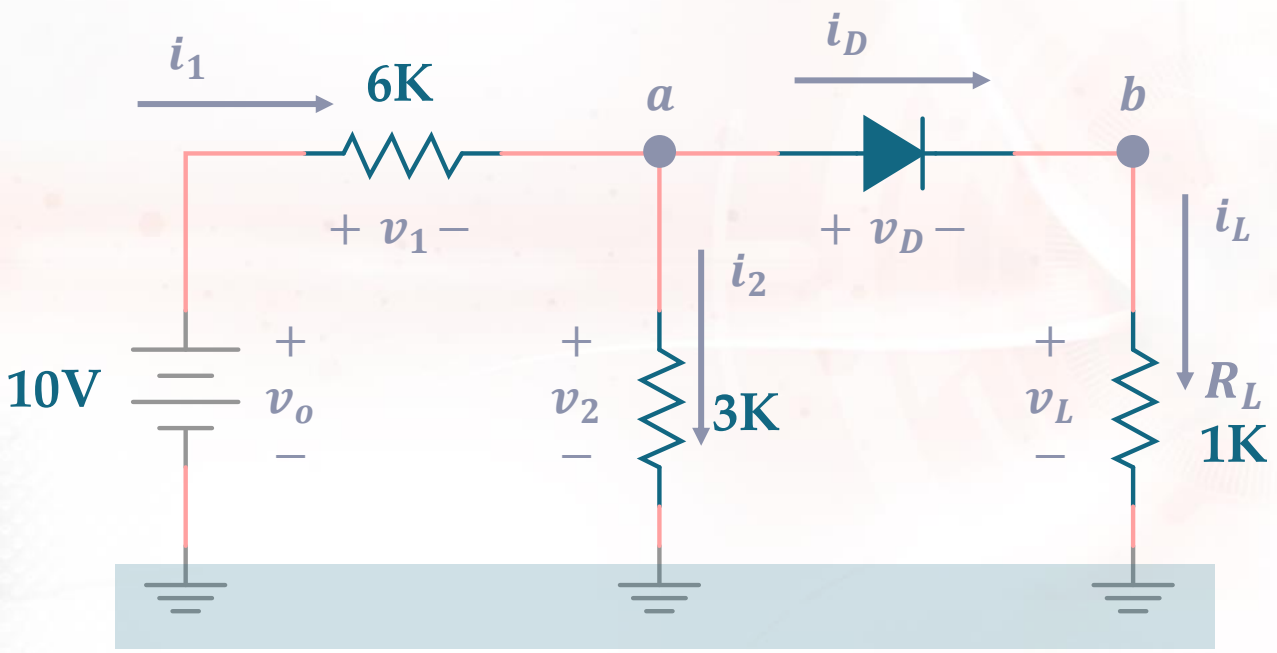
$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_L}{R_L} = 0$$

$$-\frac{v_o - v_a}{R_1} + \frac{v_a - 0}{R_2} + \frac{v_a - 0}{R_L} = 0$$

$$-\frac{v_o}{R_1} + \frac{v_a}{R_1} + \frac{v_a}{R_2} + \frac{v_a}{R_L} = 0$$

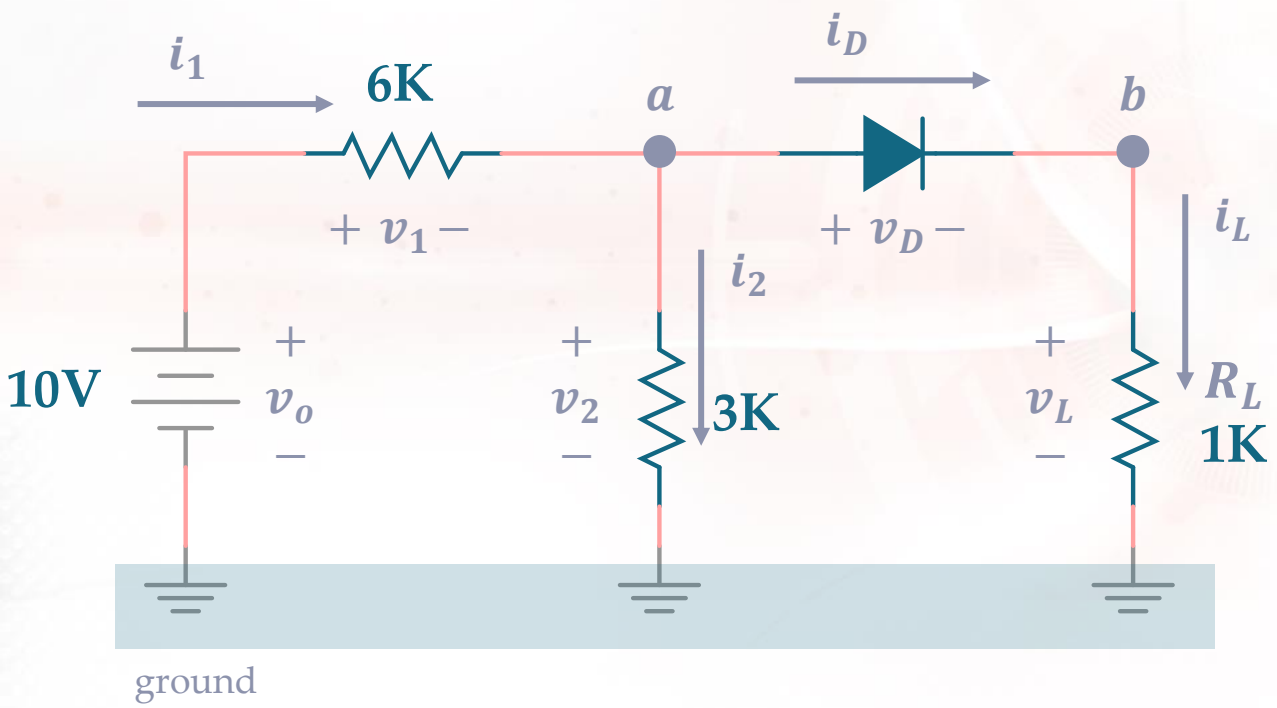
Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.

Node Analysis Method,



$$-\frac{v_o}{R_1} + \frac{v_a}{R_1} + \frac{v_a}{R_2} + \frac{v_a}{R_L} = 0$$

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.



Node Analysis Method,

$$-\frac{v_o}{R_1}+\frac{v_a}{R_1}+\frac{v_a}{R_2}+\frac{v_a}{R_L}=0$$

$$v_a\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_L}\right)=\frac{v_o}{R_1}$$

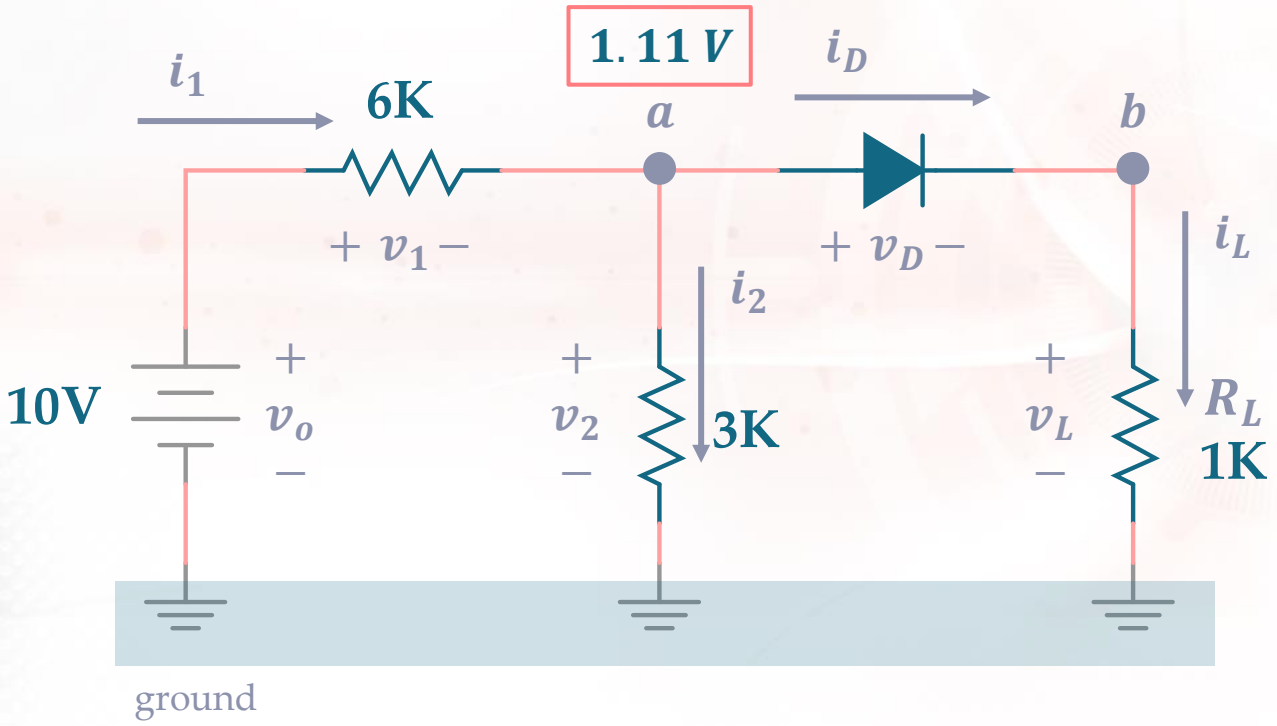
$$v_a\left(\frac{1}{6K}+\frac{1}{3K}+\frac{1}{1K}\right)=\frac{10}{6K}$$

$$0.0015v_a=\frac{1}{600}$$

$$v_a=\frac{1}{0.0015(600)}$$

$$v_a=1.11\,V$$

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.



Node Analysis Method,

$$\boxed{v_L = 1.11\ V}$$

ans

$$i_L = \frac{v_L}{R_L}$$

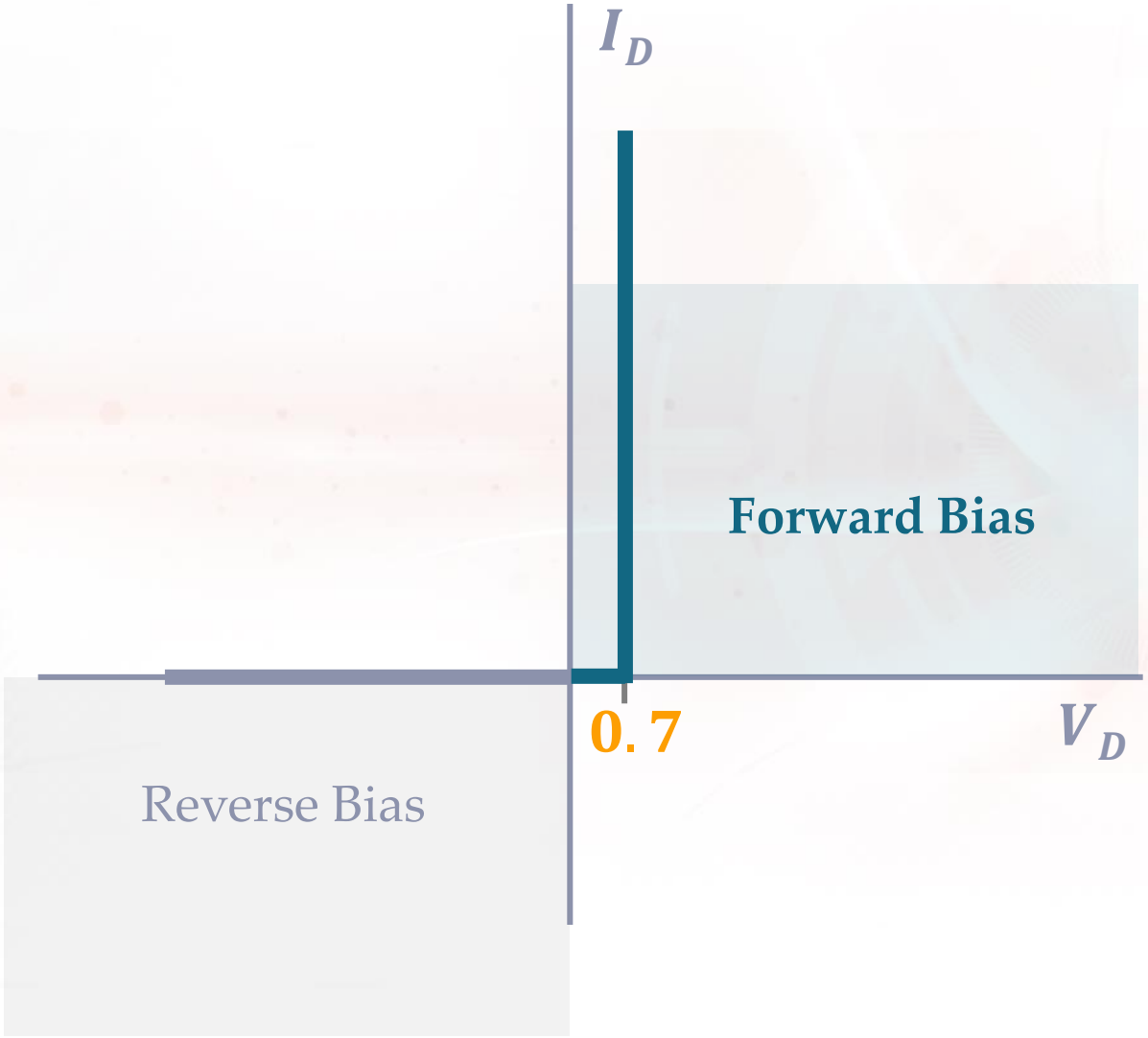
$$i_L = \frac{1.11}{1K}$$

$$\boxed{i_L = 1.11\ mA}$$

ans

2ND APPROXIMATION

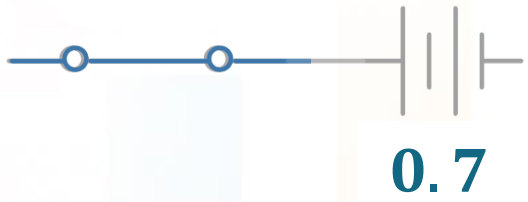
Characteristic Curve



Switch in series with a **barrier potential**

If the $\boldsymbol{V_D \geq 0.7}$, the switch is **close**

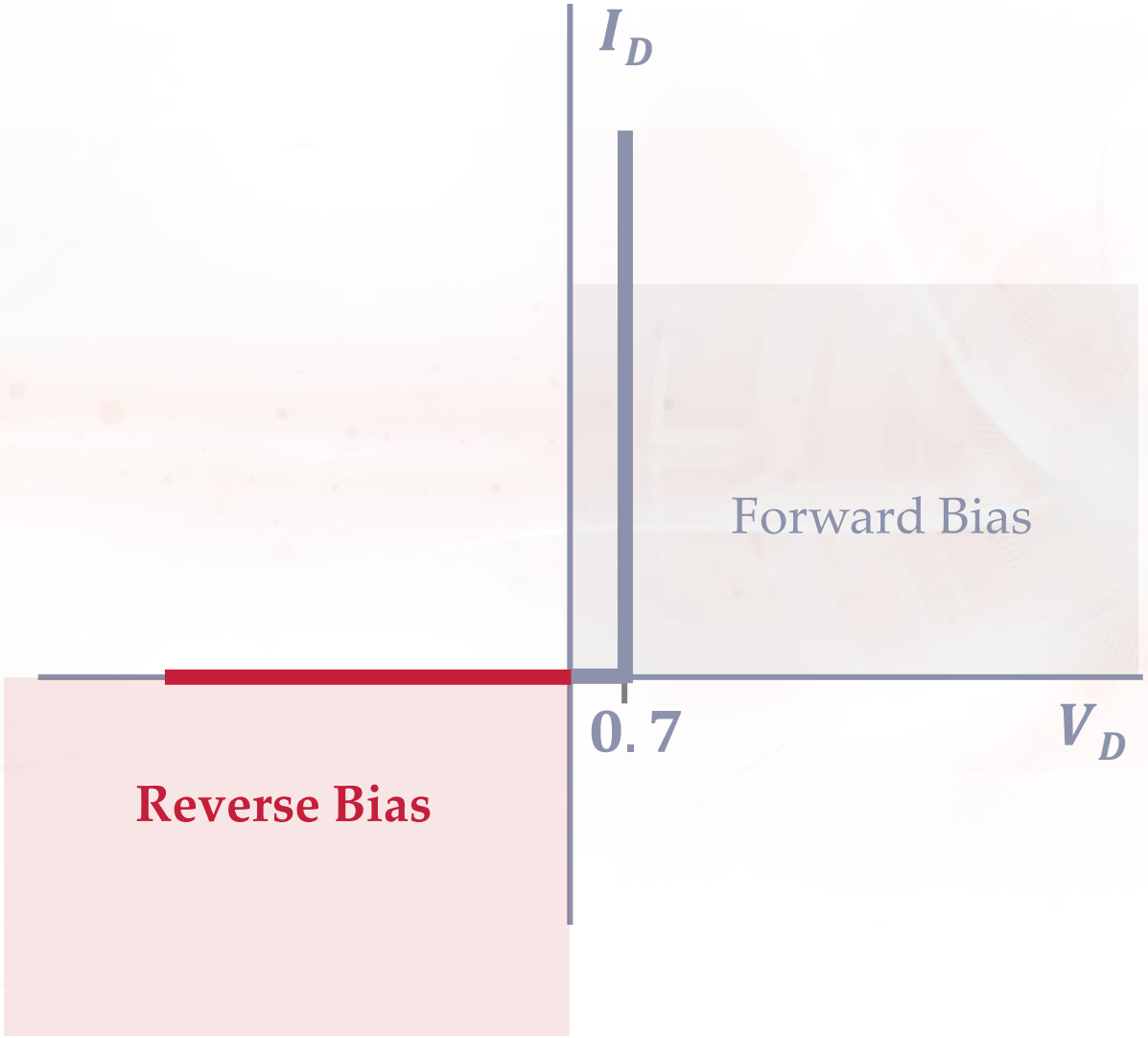
Circuit Equivalent



Forward Bias

2ND APPROXIMATION

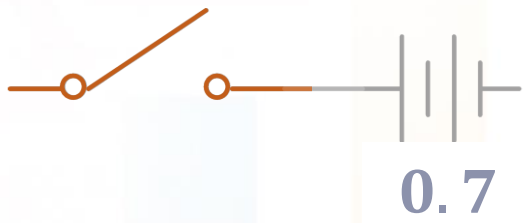
Characteristic Curve



Switch in series with a **barrier potential**:

If the $\boldsymbol{V_D < 0.7}$, the switch is **open**

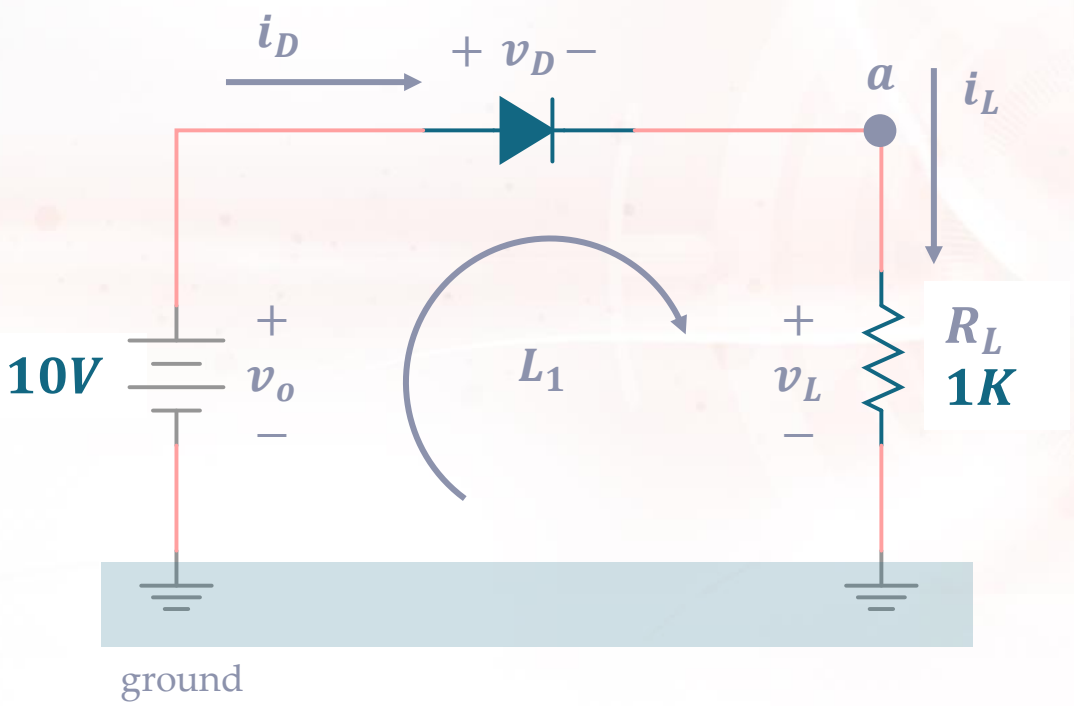
Circuit Equivalent



Reverse Bias

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



KVL @ L_1,

$$-v_0 + v_D + v_L = 0$$

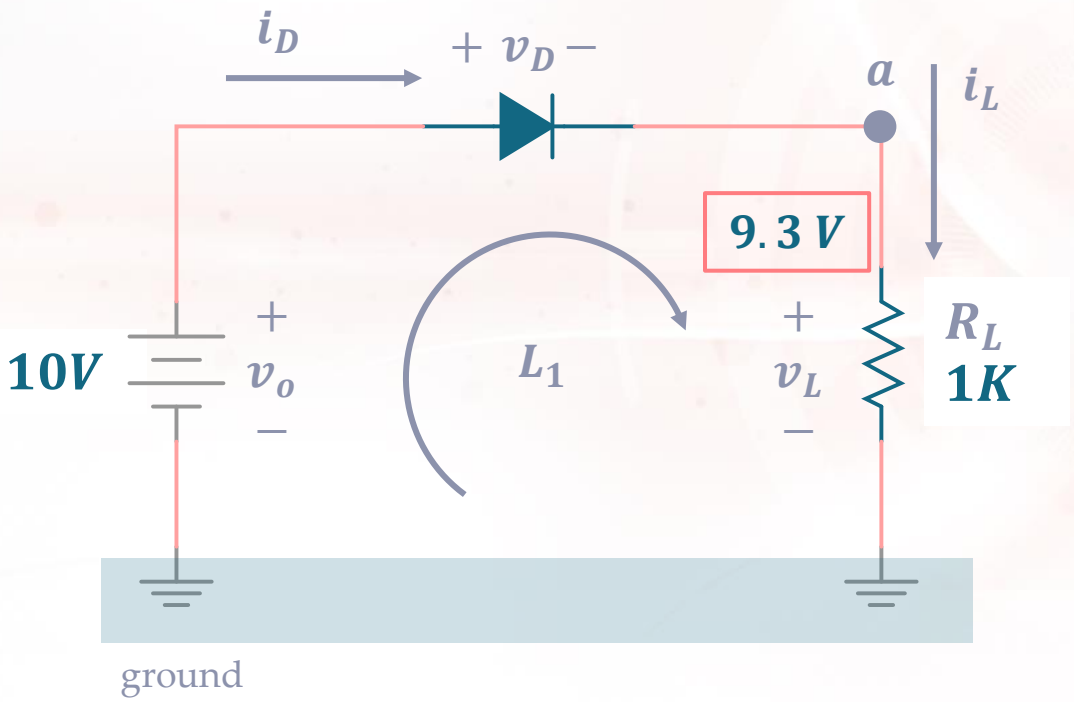
$$v_L = v_o - v_D$$

$$v_L = 10 - 0.7$$

$$\boxed{v_L = 9.3\,V}$$

ans

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



KCL @ a,

$$-i_D + i_L = 0$$

$$i_D = i_L$$

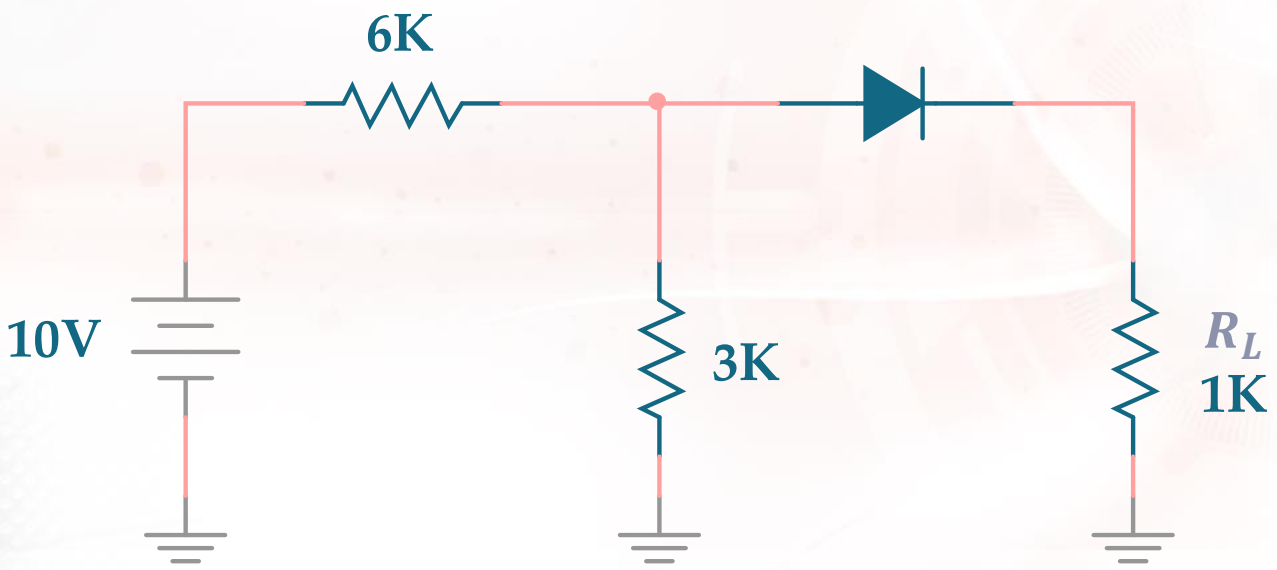
$$i_D = \frac{v_L}{R_L}$$

$$i_D = \frac{9.3}{1K}$$

$$\boxed{i_L = 9.3\,mA}$$

ans

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



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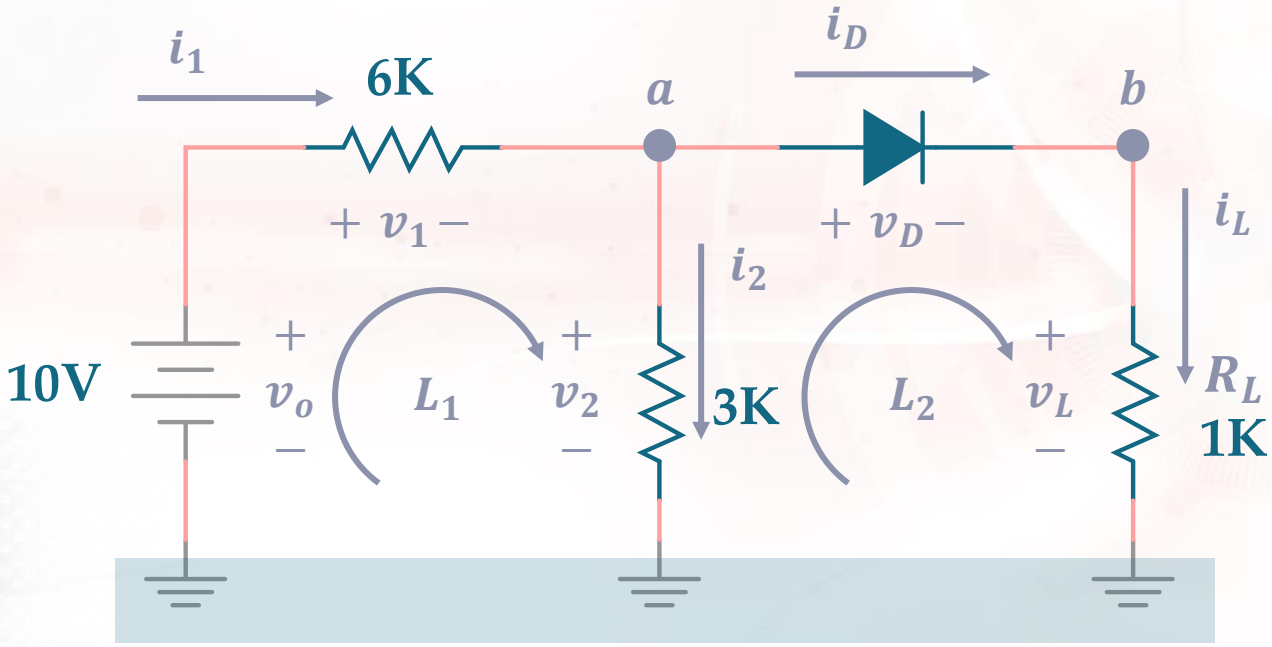


KCL @ a,

$$-i_1 + i_2 + i_D = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + i_D = 0$$

KCL @ b,

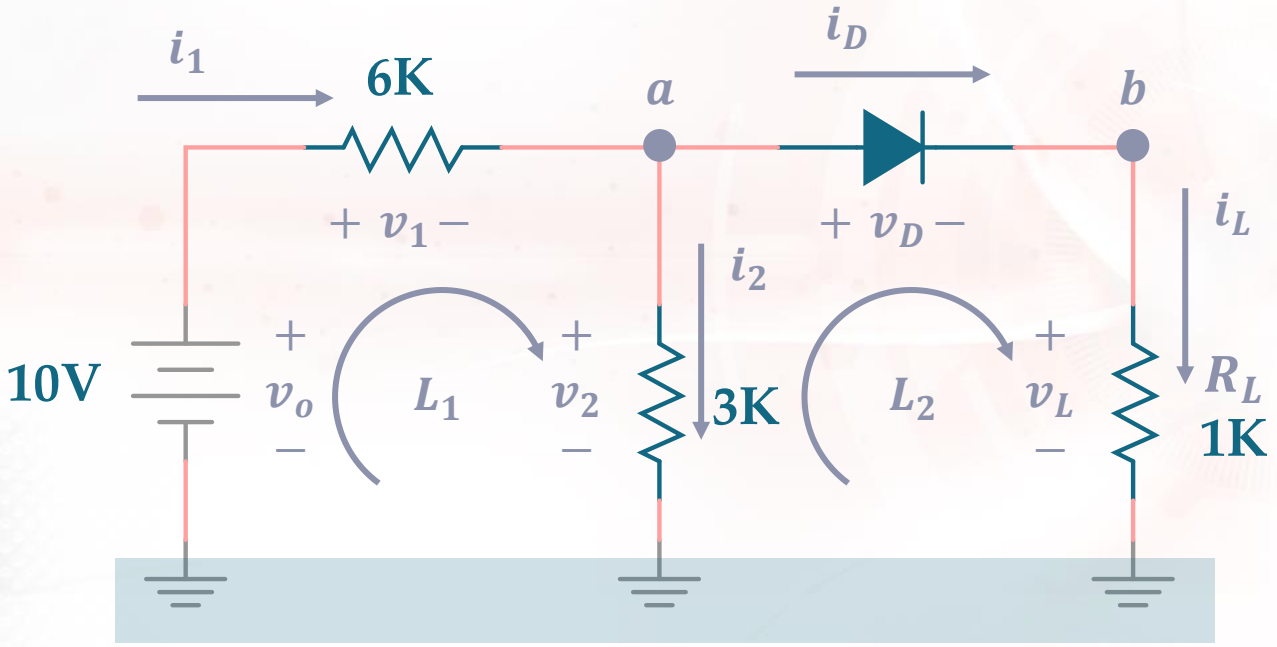
$$-i_D + i_L = 0$$

$$i_D = i_L$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + i_L = 0$$

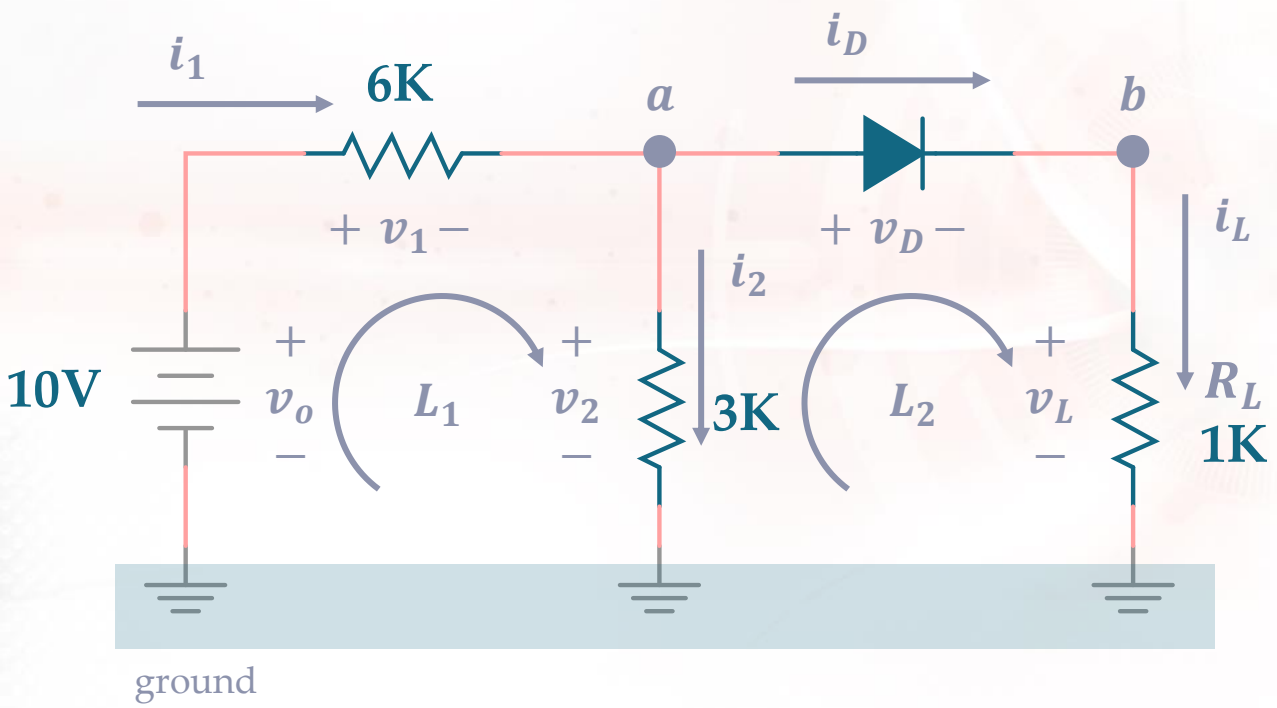
$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_L}{R_L} = 0$$

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_L}{R_L} = 0$$

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



$$-\frac{v_1}{R_1}+\frac{v_2}{R_2}+\frac{v_L}{R_L}=0$$

1 $$-\frac{v_1}{6K}+\frac{v_2}{3K}+\frac{v_L}{1K}=0$$

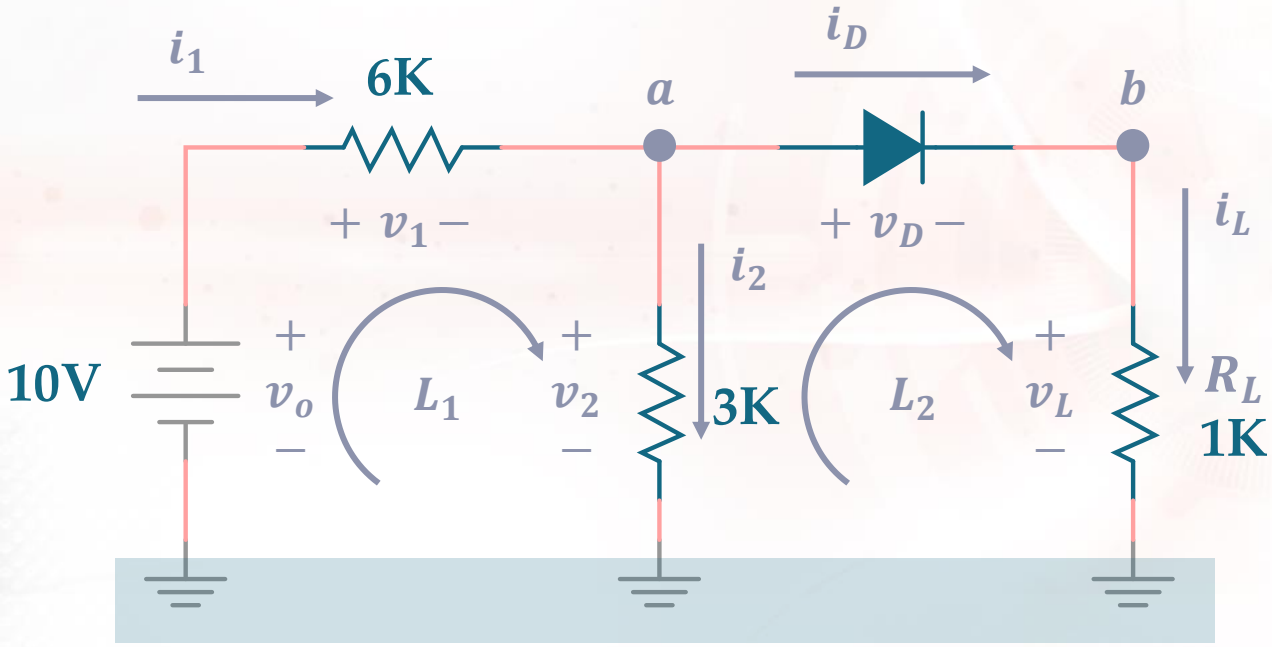
KVL @ L_1,

$$-v_o+v_1+v_2=0$$

$$-10+v_1+v_2=0$$

2 $$v_1+v_2=10$$

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



ground

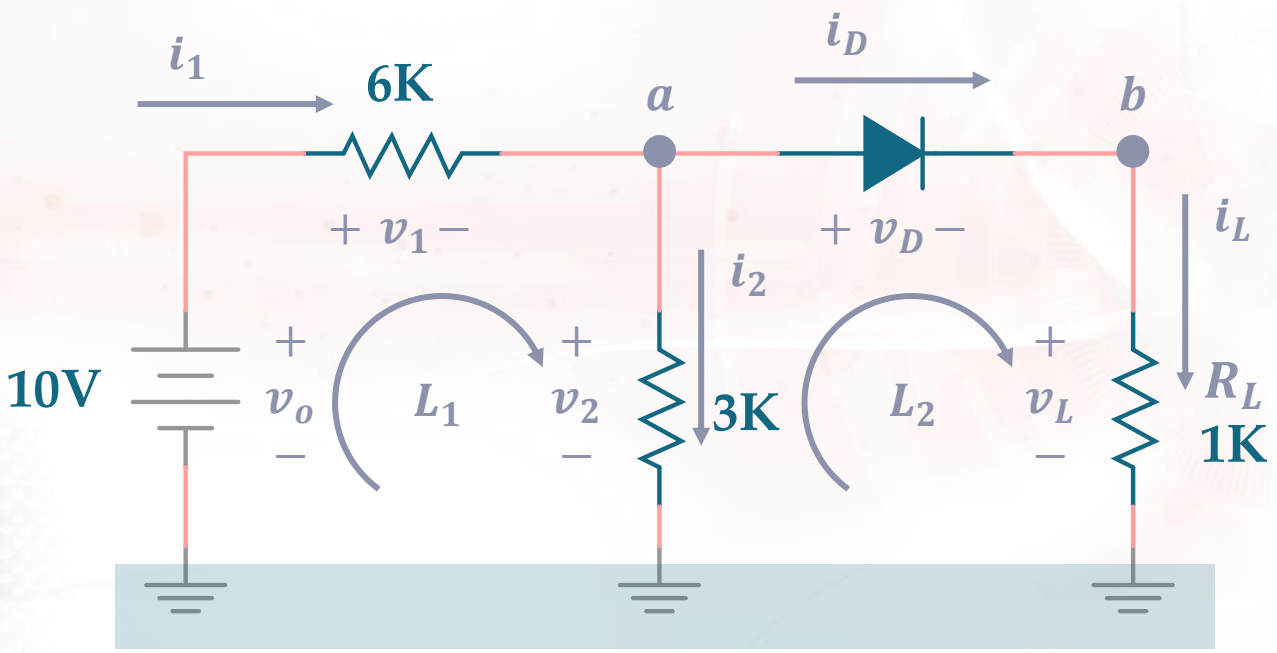
KVL @ L_2,

$$-v_2 + v_D + v_L = 0$$

$$-v_2 + 0.7 + v_L = 0$$

3 $$v_2 - v_L = 0.7$$

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



Linear Equations,

$$-\frac{v_1}{6K} + \frac{v_2}{3K} + \frac{v_L}{1K} = 0$$

$$v_1 + v_2 = 10$$

$$v_2 - v_L = 0.7$$

Augmented Matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} \\ & \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 1 & 1 & 0 & 10 \\ 0 & 1 & -1 & 0.7 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

Linear Equations,

$$-\frac{v_1}{6K} + \frac{v_2}{3K} + \frac{v_L}{1K} = 0$$

$$v_1 + v_2 = 10$$

$$v_2 - v_L = 0.7$$

Augmented Matrix,

$$
\begin{array}{cc}
\begin{array}{cccc} v_1 & v_2 & v_L & \end{array} & \\
\begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 1 & 1 & 0 & 10 \\ 0 & 1 & -1 & 0.7 \end{bmatrix} & \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}
\end{array}
$$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 1 & 1 & 0 & 10 \\ 0 & 1 & -1 & 0.7 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 0 & 3 & 6 & 10 \\ 0 & 1 & -1 & 0.7 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$r_2 \rightarrow 6Kr_1 + r_2$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 0 & 3 & 6 & 10 \\ 0 & 0 & 9 & 7.9 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 0 & 3 & 6 & 10 \\ 0 & 1 & -1 & 0.7 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$r_3 \rightarrow -3r_3 + r_2$$

Augmented Matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_L & \end{array} \\ & \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{1K} & 0 \\ 0 & 3 & 6 & 10 \\ 0 & 0 & 9 & 7.9 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

from equation 3,

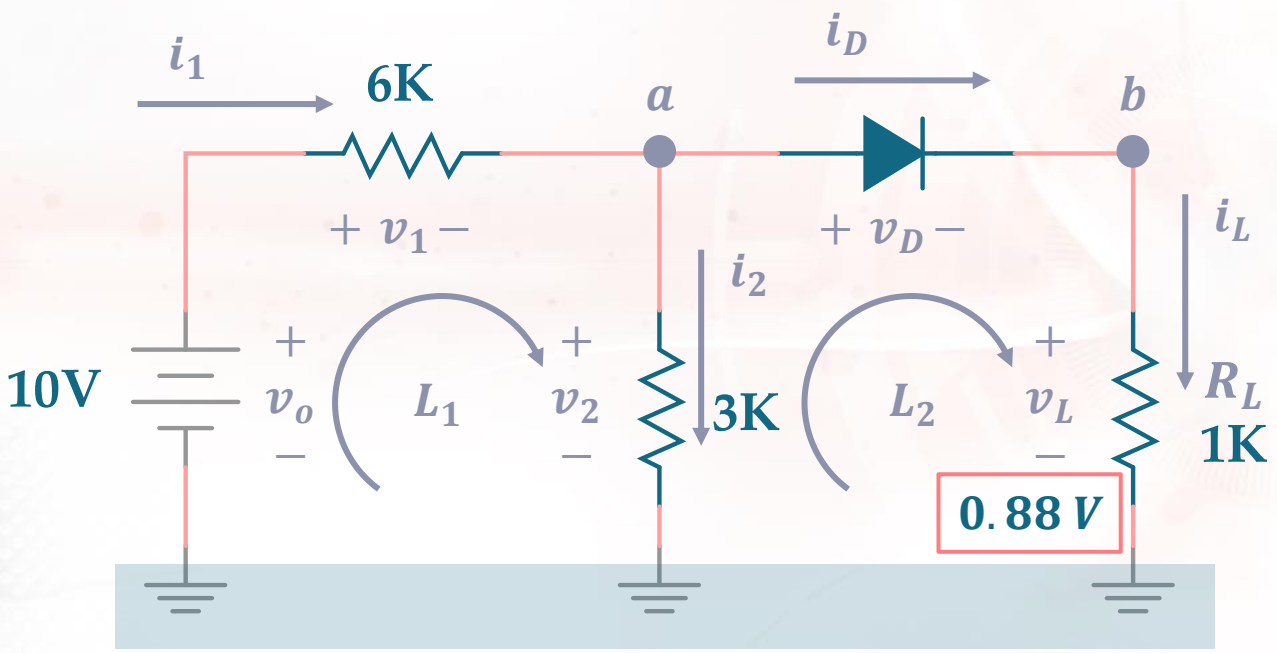
$$9v_L = 7.9$$

$$v_L = \frac{7.9}{9}$$

$$\boxed{v_L = 0.88\ V}$$

ans

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



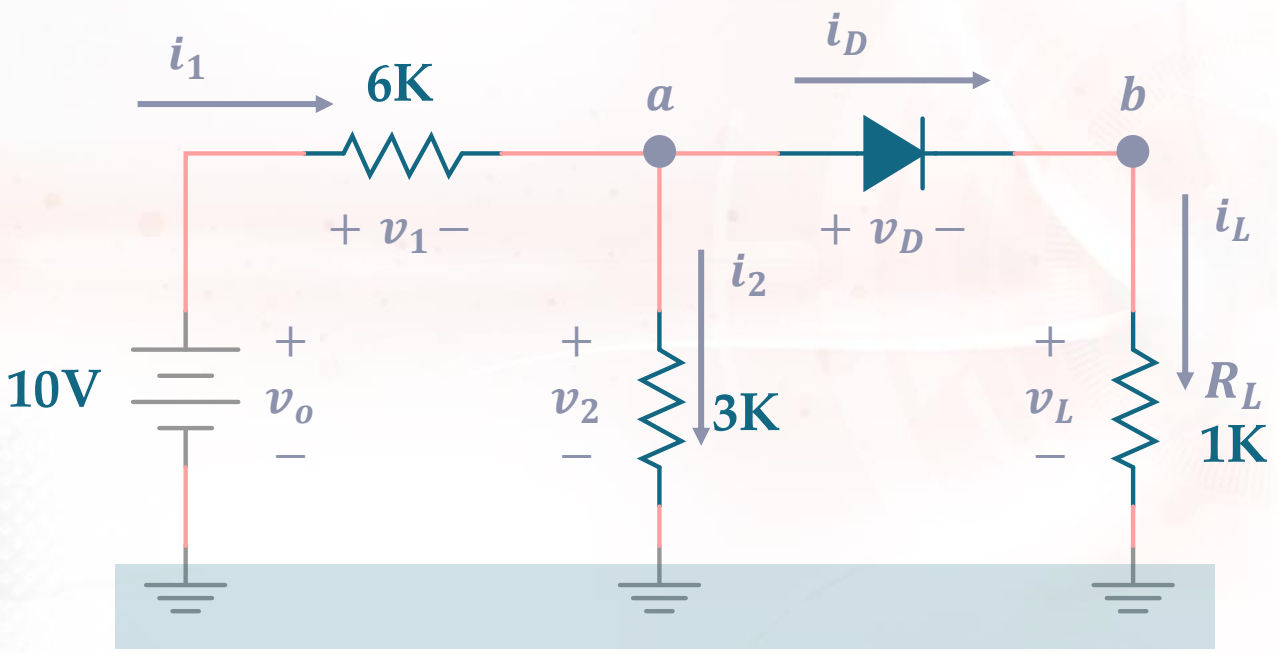
$$i_L = \frac{v_L}{R_L}$$

$$i_L = \frac{0.88}{1K}$$

$$\boxed{i_L = 880\ \mu A}$$

ans

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



Node Analysis Method,

KCL @ a,

$$-i_1 + i_2 + i_D = 0$$

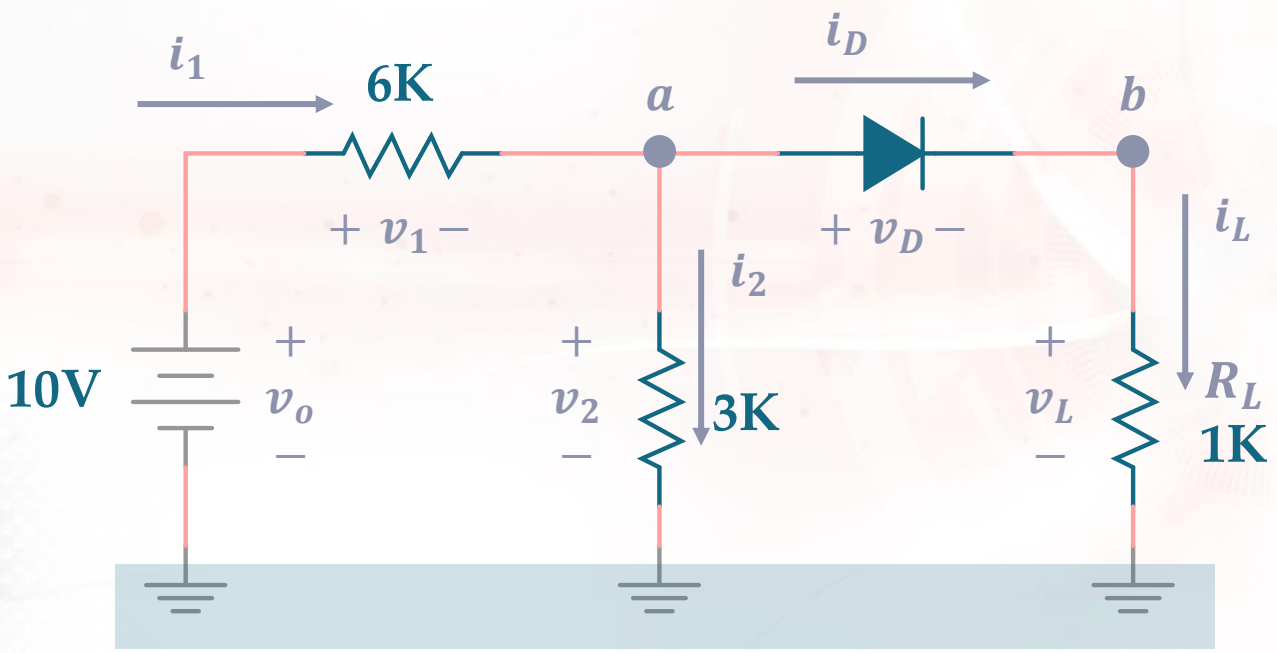
$$-i_1 + i_2 + i_L = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_L}{R_L} = 0$$

$$-\frac{v_o - v_a}{R_1} + \frac{v_a - 0}{R_2} + \frac{v_b - 0}{R_L} = 0$$

$$-\frac{v_o}{R_1} + \frac{v_a}{R_1} + \frac{v_a}{R_2} + \frac{v_b}{R_L} = 0$$

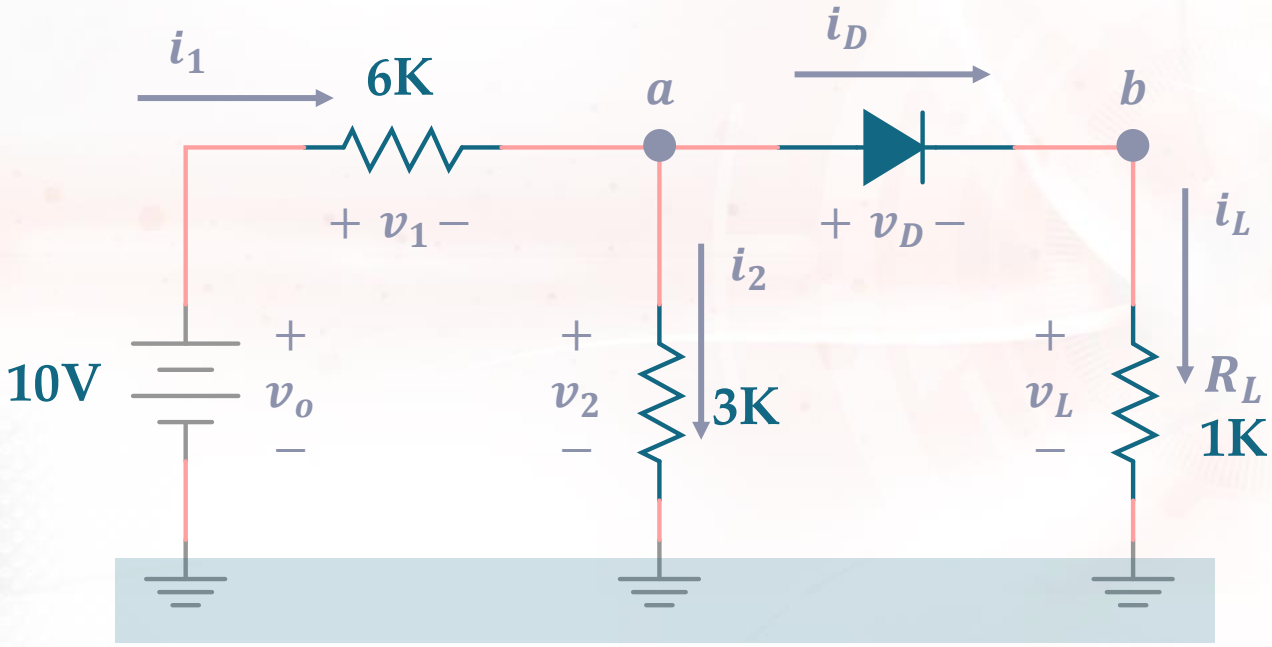
Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



Node Analysis Method,

$$-\frac{v_o}{R_1}+\frac{v_a}{R_1}+\frac{v_a}{R_2}+\frac{v_b}{R_L}=0$$

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



Node Analysis Method,

$$-\frac{v_o}{R_1}+\frac{v_a}{R_1}+\frac{v_a}{R_2}+\frac{v_b}{R_L}=0$$

$$v_D = v_a - v_b$$

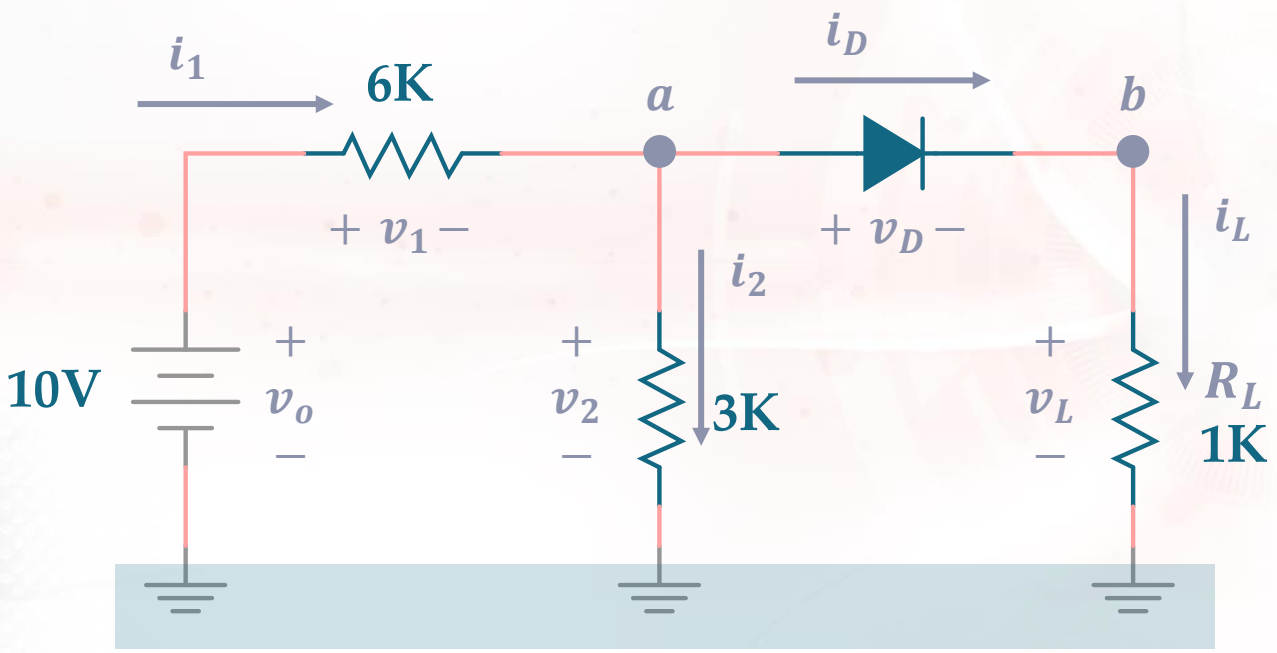
$$0.7 = v_a - v_b$$

$$v_b = v_a - 0.7$$

$$-\frac{v_o}{R_1}+\frac{v_a}{R_1}+\frac{v_a}{R_2}+\frac{v_a-0.7}{R_L}=0$$

$$-\frac{v_o}{R_1}+\frac{v_a}{R_1}+\frac{v_a}{R_2}+\frac{v_a}{R_L}-\frac{0.7}{R_L}=0$$

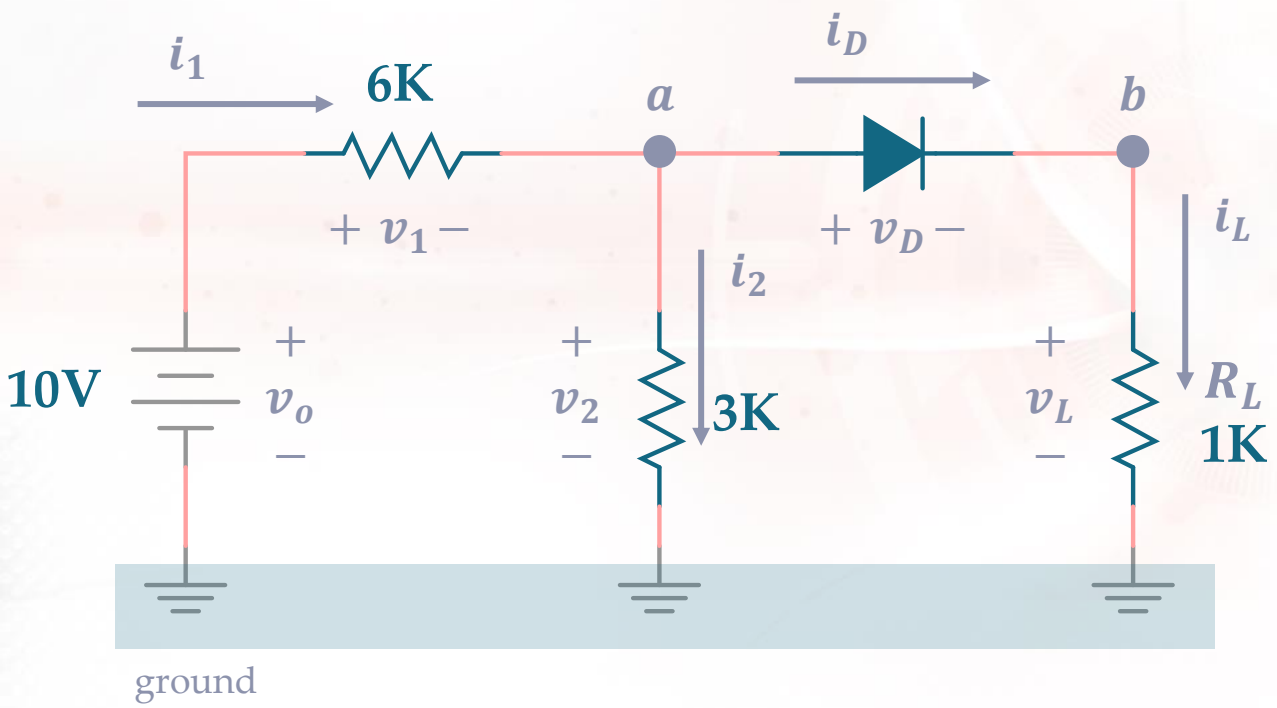
Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



Node Analysis Method,

$$-\frac{v_o}{R_1}+\frac{v_a}{R_1}+\frac{v_a}{R_2}+\frac{v_a}{R_L}-\frac{0.7}{R_L}=0$$

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



Node Analysis Method,

$$-\frac{v_o}{R_1}+\frac{v_a}{R_1}+\frac{v_a}{R_2}+\frac{v_a}{R_L}-\frac{0.7}{R_L}=0$$

$$v_a\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_L}\right)=\frac{v_o}{R_1}+\frac{0.7}{R_L}$$

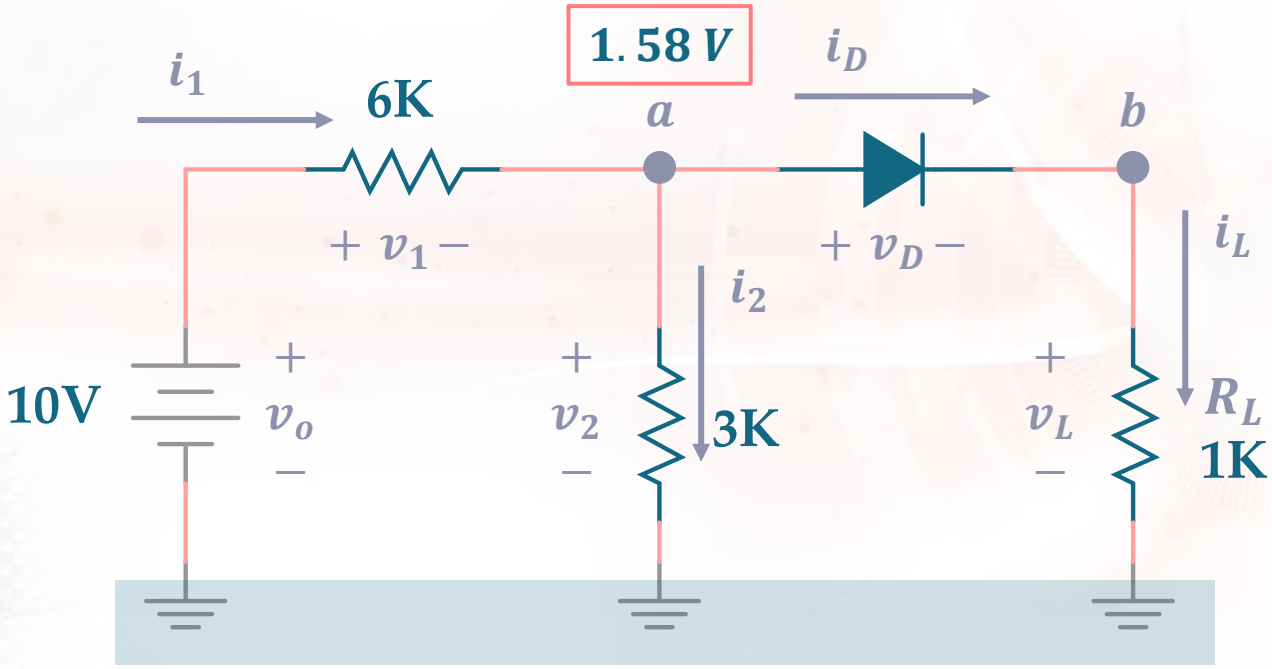
$$v_a\left(\frac{1}{6K}+\frac{1}{3K}+\frac{1}{1K}\right)=\frac{10}{6K}+\frac{0.7}{1K}$$

$$0.0015v_a=\frac{71}{30K}$$

$$v_a=\frac{71}{30K(0.0015)}$$

$$v_a=1.58\,V$$

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



Node Analysis Method,

$$v_D = v_a - v_b$$

$$v_b = v_a - v_D$$

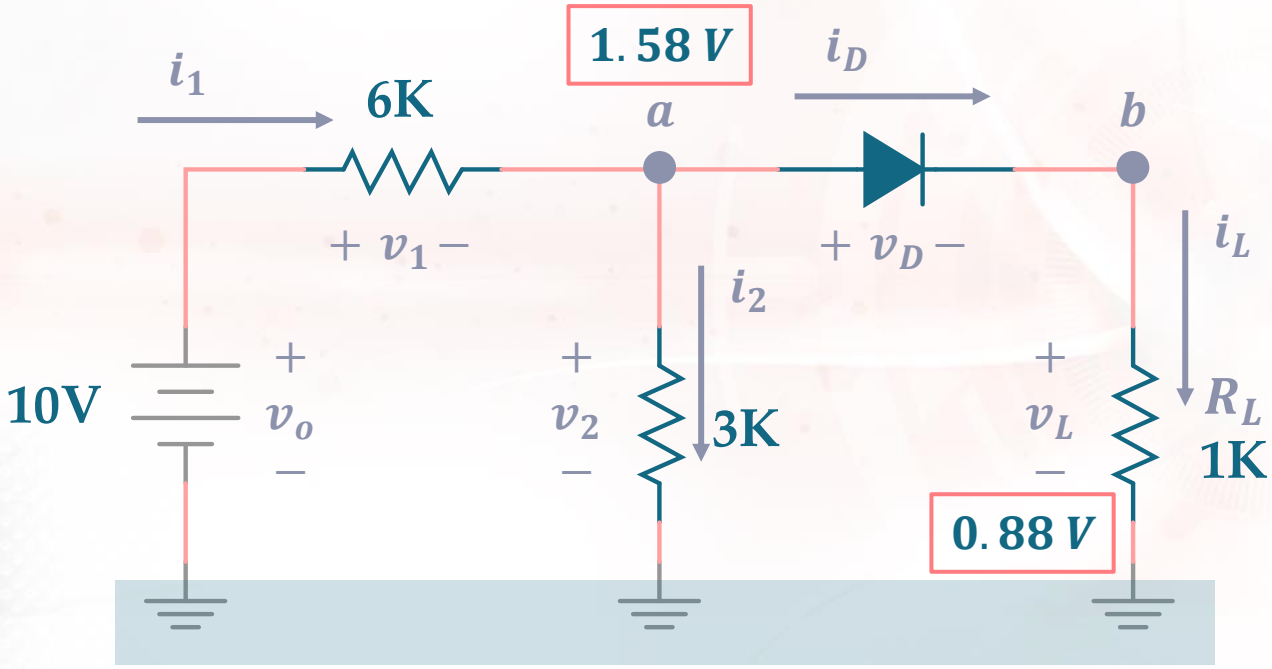
$$v_b = 1.58 - 0.7$$

$$v_b = 0.88\ V$$

$$\boxed{v_L = 0.88\ V}$$

ans

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



Node Analysis Method,

$$i_L = \frac{v_L}{R_L}$$

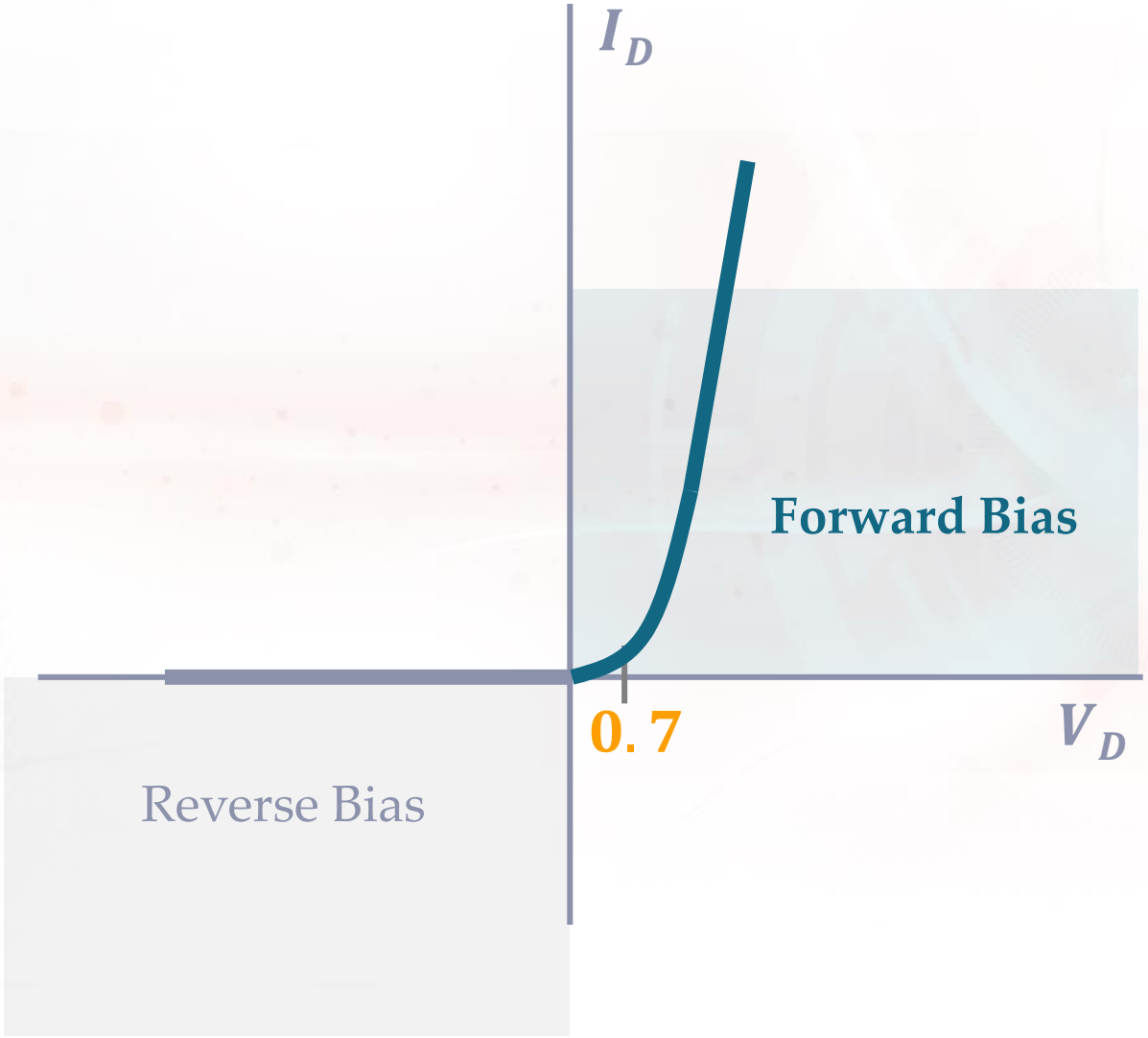
$$i_L = \frac{0.88}{1K}$$

$$\boxed{i_L = 880\ \mu A}$$

ans

3RD APPROXIMATION

Characteristic Curve

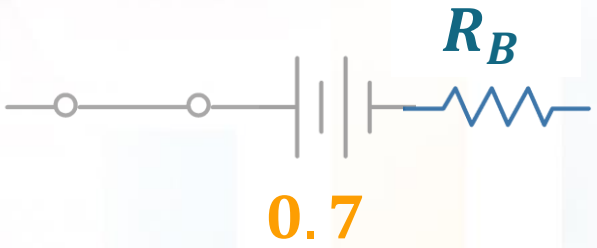


Bulk Resistance is included:

$$V_D = 0.7V + I_D R_B$$

Ignore bulk: $R_B < 0.01 R_{TH}$

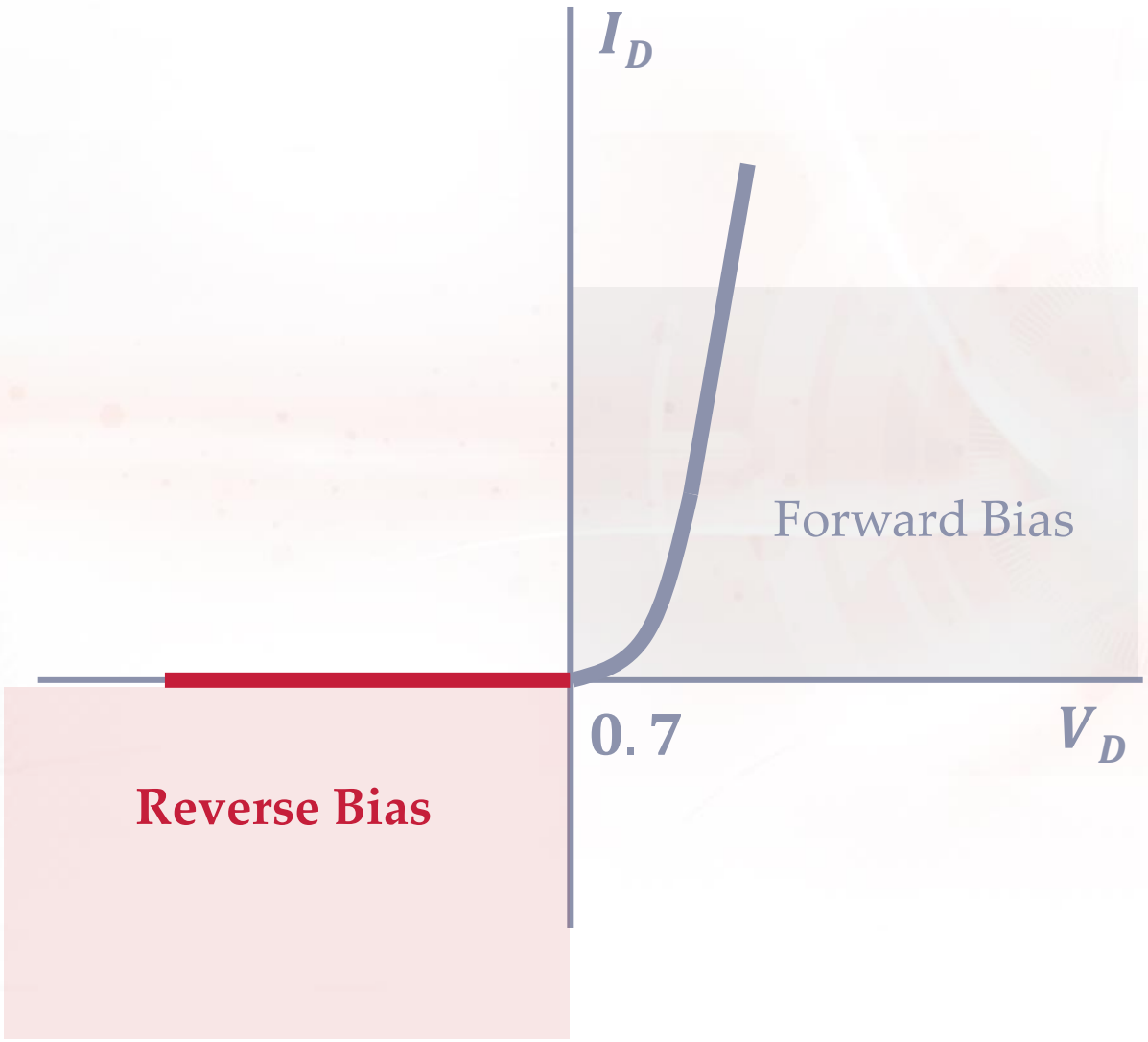
Circuit Equivalent



Forward Bias

3RD APPROXIMATION

Characteristic Curve



Bulk Resistance is included:

$V_D = 0.7V + I_D R_B$

Ignore bulk: $R_B < 0.01 R_{TH}$

Circuit Equivalent

Reverse Bias

BULK RESISTANCE

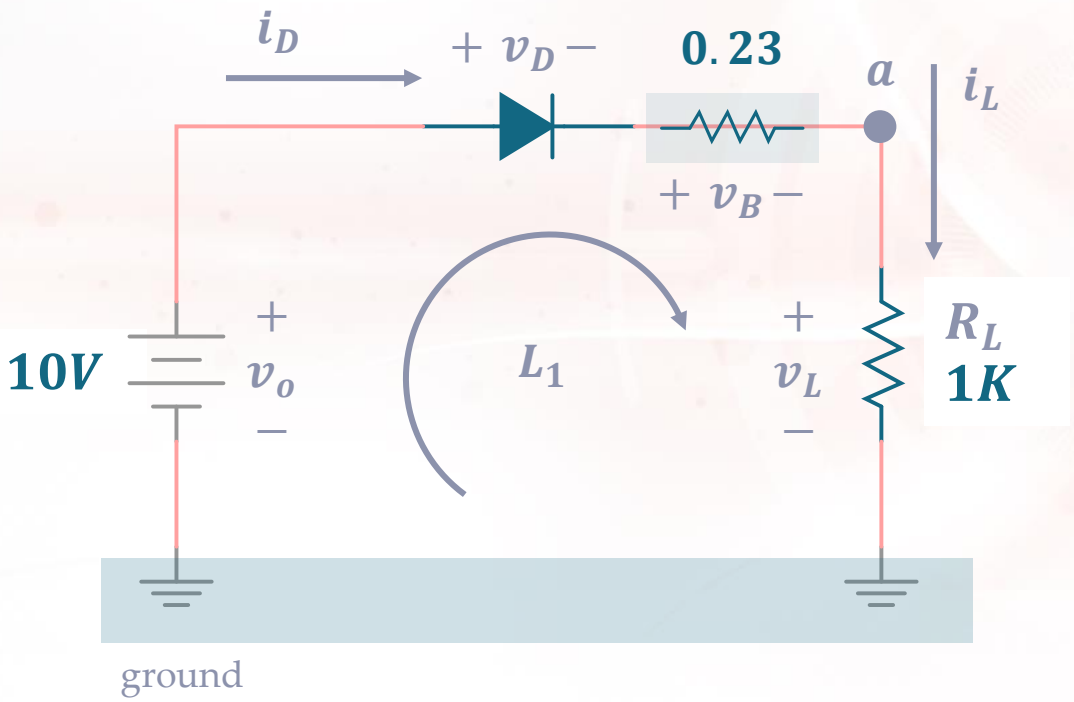
The sum of ohmic resistance of $\boldsymbol{p}$ and $\boldsymbol{n}$ region

$$\boldsymbol{R_B = R_P + R_N}$$

$$\boldsymbol{R_B < 1\Omega} \quad (typ)$$

The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?

The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?



KVL @ L_1,

$$-v_0 + v_D + v_B + v_L = 0$$

$$v_B + v_L = v_o - v_D$$

$$v_B + v_L = 10 - 0.7$$

1 $$v_B + v_L = 9.3$$

KCL @ a,

$$-i_D + i_L = 0$$

$$-\frac{v_B}{R_B} + \frac{v_L}{R_L} = 0$$

2 $$-\frac{v_B}{0.23} + \frac{v_L}{1K} = 0$$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{ccc} v_B & v_L & \end{array} \\ \left[\begin{array}{ccc} 1 & 1 & 9.3 \\ -\dfrac{1}{0.23} & \dfrac{1}{1K} & 0 \end{array}\right] \end{array} \begin{array}{c} \\ r_1 \\ r_2 \end{array}$$

Linear Equations,

$$v_B + v_L = 9.3$$

$$-\frac{v_B}{0.23} + \frac{v_L}{1K} = 0$$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{cc} v_B & v_L \end{array} \\ \left[\begin{array}{cc|c} 1 & 1 & 9.3 \\ -\dfrac{1}{0.23} & \dfrac{1}{1K} & 0 \end{array}\right] \end{array} \begin{array}{c} r_1 \\ r_2 \end{array}$$

$$\begin{array}{c} \begin{array}{cc} v_B & v_L \end{array} \\ \left[\begin{array}{cc|c} 1 & 1 & 9.3 \\ 0 & 1.00023 & 9.3 \end{array}\right] \end{array} \begin{array}{c} r_1 \\ r_2 \end{array}$$

$r_2 \rightarrow 0.23r_2 + r_1$

from equation 2,

$$1.00023v_L = 9.3$$

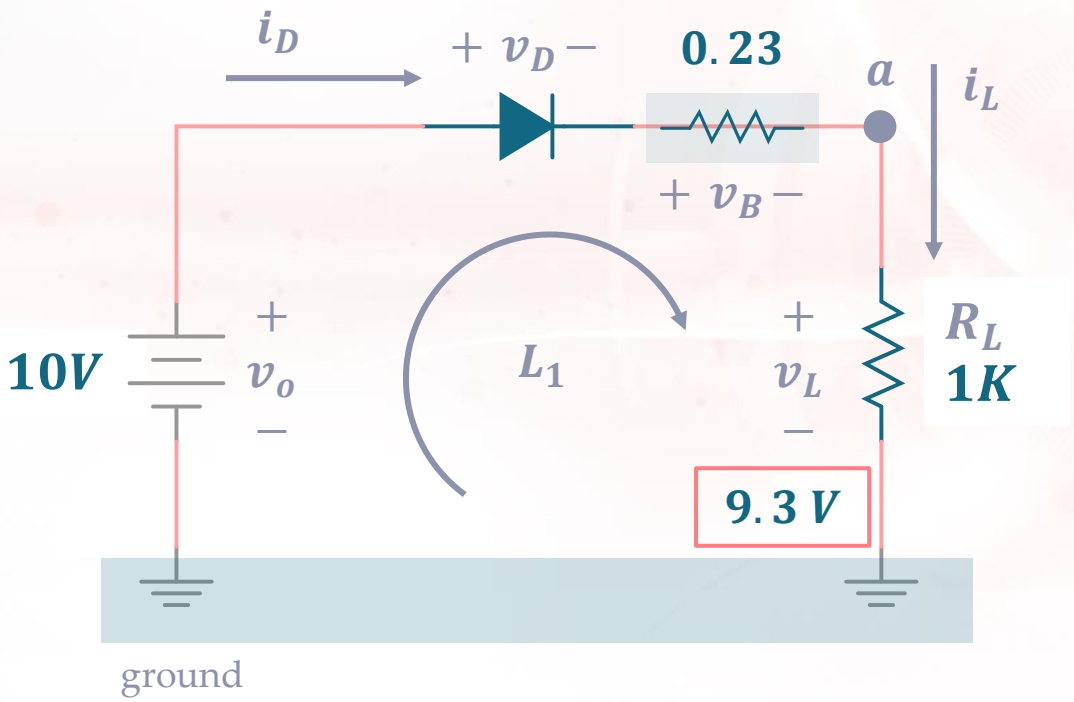
$$v_L = \frac{9.3}{1.00023}$$

$$\boxed{v_L = 9.3\ V}$$

ans

$$\begin{array}{cc} & \begin{array}{ccc} v_B & v_L & \end{array} \\ & \begin{bmatrix} 1 & 1 & 9.3 \\ 0 & 1.00023 & 9.3 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \end{array} \end{array}$$

The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?



$$i_L = \frac{v_L}{R_L}$$

$$i_L = \frac{9.3}{1K}$$

$$\boxed{i_L = 9.3\ mA}$$

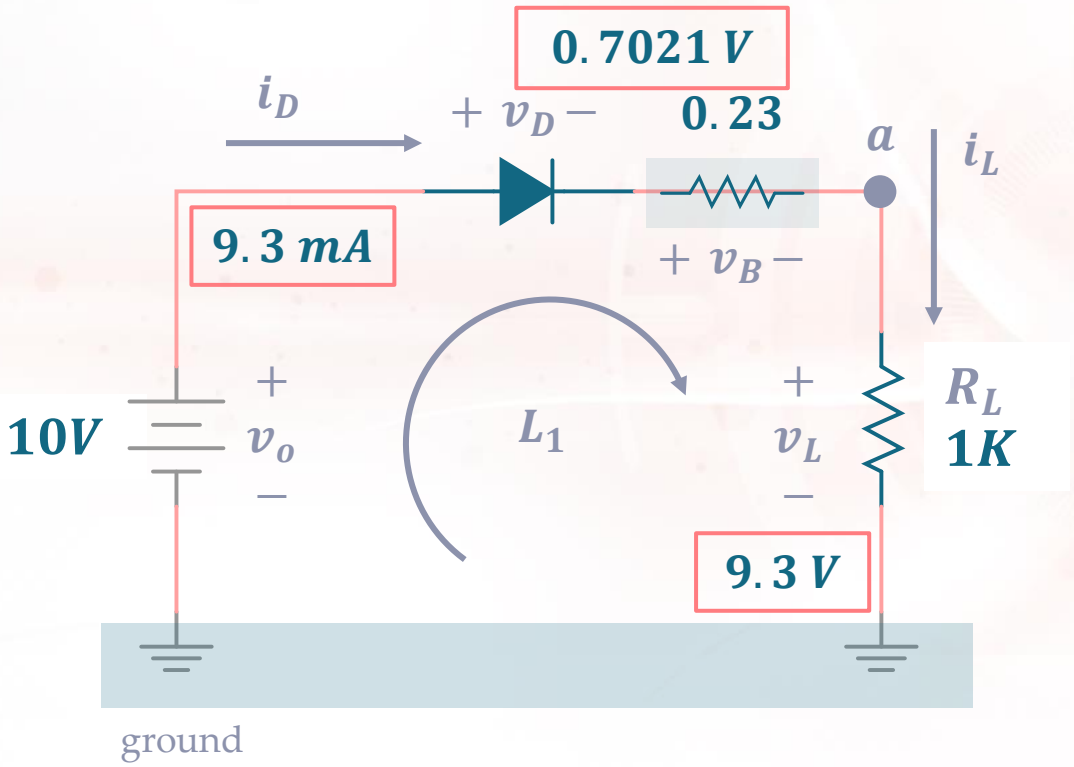
ans

$$V_D = 0.7 + i_D R_B$$

$$V_D = 0.7 + 9.3mA(0.23)$$

$$V_D = 0.7021$$

The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?



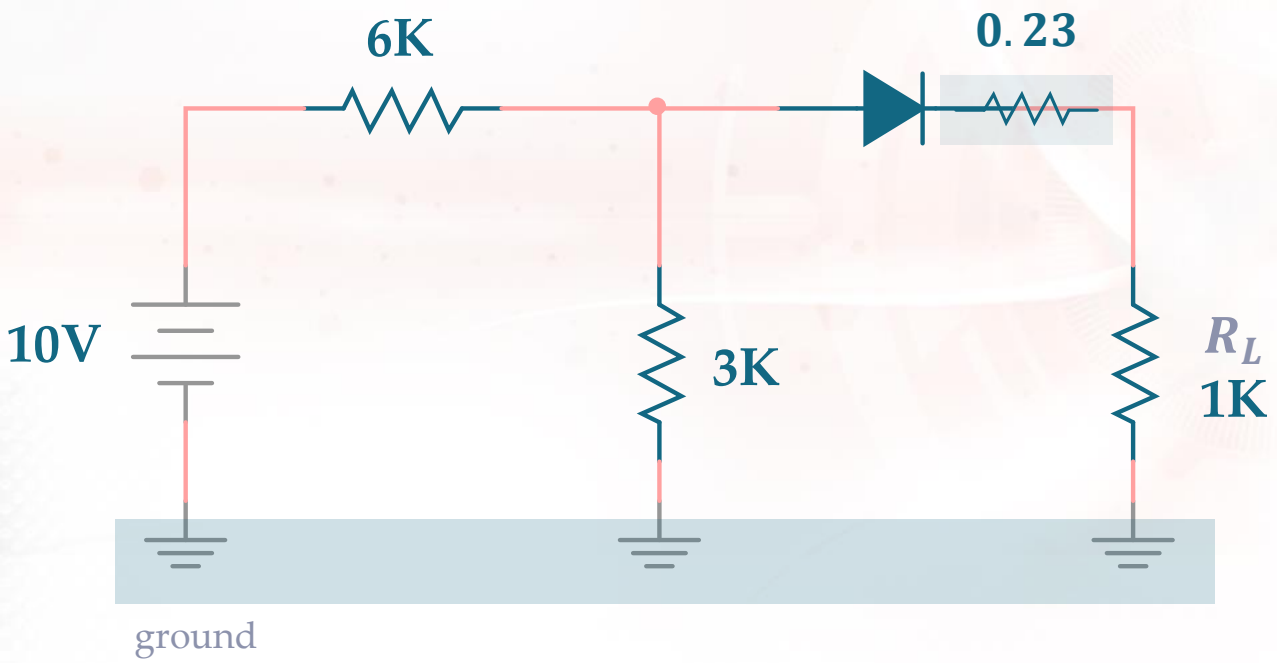
$$P_D = i_D v_D$$

$$P_D = 9.3\ mA(0.7021)$$

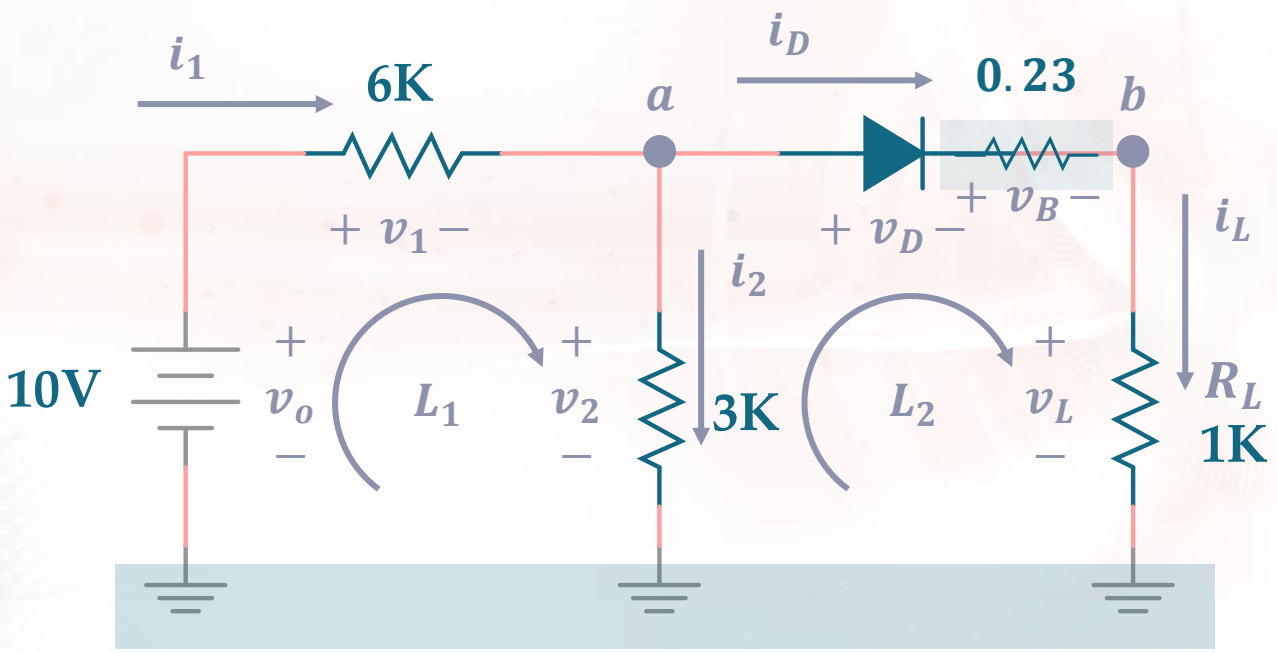
$$\boxed{P_D = 6.53\ mW}$$

ans

The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?



The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?



KCL @ a,

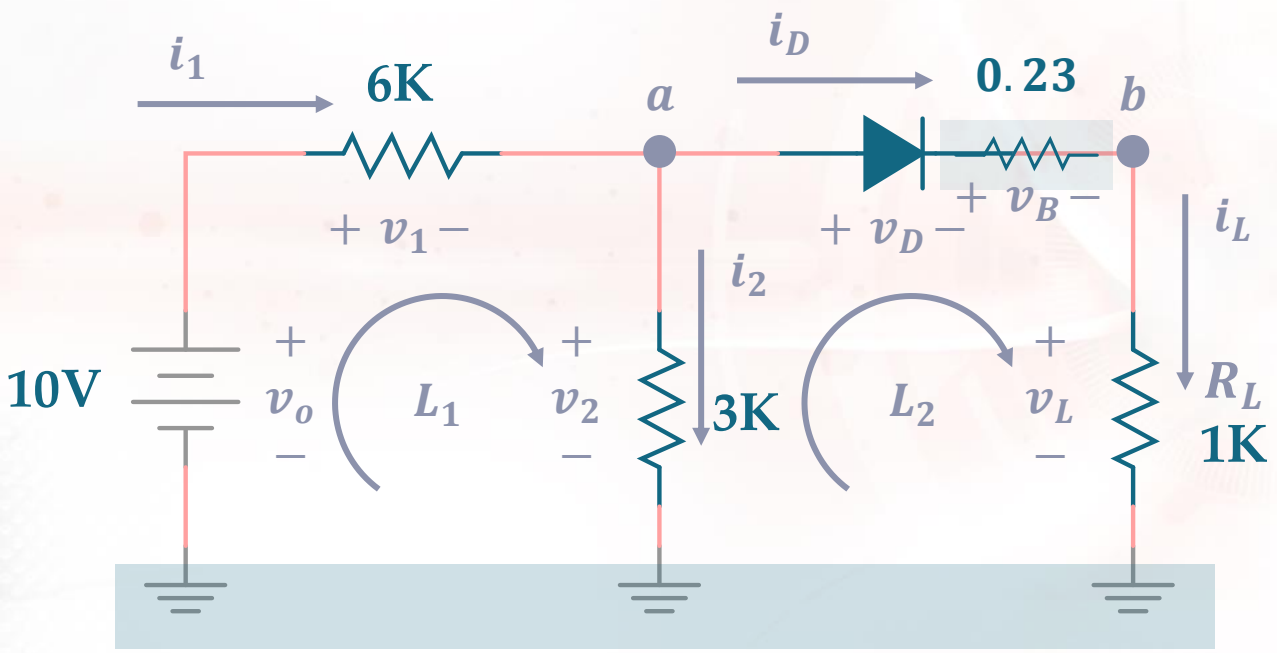
$$-i_1 + i_2 + i_D = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + i_D = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_B}{R_B} = 0$$

1 $$-\frac{v_1}{6K} + \frac{v_2}{3K} + \frac{v_B}{0.23} = 0$$

The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?



ground

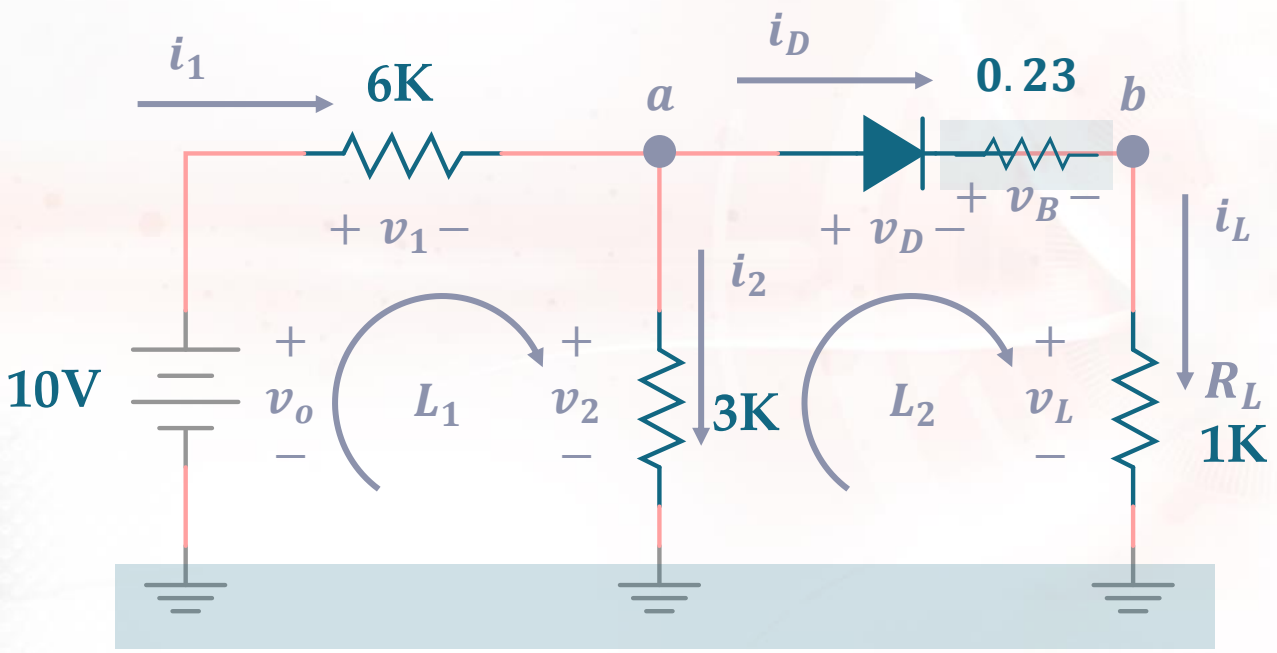
KCL @ b,

$$-i_D + i_L = 0$$

$$-\frac{v_B}{R_B} + \frac{v_L}{R_L} = 0$$

2 $$-\frac{v_B}{0.23} + \frac{v_L}{1K} = 0$$

The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?



KVL @ L_1,

$$-v_o + v_1 + v_2 = 0$$

(3) $v_1 + v_2 = 10$

KVL @ L_2,

$$-v_2 + v_D + v_B + v_L = 0$$

$$-v_2 + 0.7 + v_B + v_L = 0$$

(4) $v_2 - v_B - v_L = 0.7$

Augmented Matrix,

$$
\begin{array}{cc}
\begin{array}{ccccc} v_1 & v_2 & v_B & v_L & \end{array} & \\
\begin{bmatrix}
-\frac{1}{6K} & \frac{1}{3K} & \frac{1}{0.23} & 0 & 0 \\
0 & 0 & -\frac{1}{0.23} & \frac{1}{1K} & 0 \\
1 & 1 & 0 & 0 & 10 \\
0 & 1 & -1 & -1 & 0.7
\end{bmatrix} &
\begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \end{array}
\end{array}
$$

Linear Equations,

$$-\frac{v_1}{6K} + \frac{v_2}{3K} + \frac{v_B}{0.23} = 0$$

$$-\frac{v_B}{0.23} + \frac{v_L}{1K} = 0$$

$$v_1 + v_2 = 10$$

$$v_2 - v_B - v_L = 0.7$$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_B & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{0.23} & 0 & 0 \\ 0 & 0 & -\frac{1}{0.23} & \frac{1}{1K} & 0 \\ 1 & 1 & 0 & 0 & 10 \\ 0 & 1 & -1 & -1 & 0.7 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \end{array}$$

$$\begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_B & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{0.23} & 0 & 0 \\ 0 & 1 & -1 & -1 & 0.7 \\ 0 & 0 & -\frac{1}{0.23} & \frac{1}{1K} & 0 \\ 1 & 1 & 0 & 0 & 10 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \end{array}$$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_B & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{0.23} & 0 & 0 \\ 0 & 1 & -1 & -1 & 0.7 \\ 0 & 0 & -\frac{1}{0.23} & \frac{1}{1K} & 0 \\ 0 & 3 & \frac{600K}{23} & 0 & 10 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \end{array}$$

$$\begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_B & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{0.23} & 0 & 0 \\ 0 & 1 & -1 & -1 & 0.7 \\ 0 & 0 & -\frac{1}{0.23} & \frac{1}{1K} & 0 \\ 1 & 1 & 0 & 0 & 10 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \end{array}$$

$$r_4 \rightarrow 6Kr_1 + r_4$$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_B & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{0.23} & 0 & 0 \\ 0 & 1 & -1 & -1 & 0.7 \\ 0 & 0 & -\frac{1}{0.23} & \frac{1}{1K} & 0 \\ 0 & 3 & \frac{600K}{23} & 0 & 10 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \end{array}$$

$$\begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_B & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{0.23} & 0 & 0 \\ 0 & 1 & -1 & -1 & 0.7 \\ 0 & 0 & -\frac{1}{0.23} & \frac{1}{1K} & 0 \\ 0 & 0 & \frac{600069}{23} & 3 & 7.9 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \end{array}$$

$r_4 \rightarrow -3r_2 + r_4$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_B & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{0.23} & 0 & 0 \\ 0 & 1 & -1 & -1 & 0.7 \\ 0 & 0 & -\frac{1}{0.23} & \frac{1}{1K} & 0 \\ 0 & 0 & 0 & \frac{900069}{23K} & \frac{790}{23} \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \end{array}$$

$$\begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_B & v_L & \end{array} \\ \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{0.23} & 0 & 0 \\ 0 & 1 & -1 & -1 & 0.7 \\ 0 & 0 & -\frac{1}{0.23} & \frac{1}{1K} & 0 \\ 0 & 0 & \frac{600069}{23} & 3 & 7.9 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \end{array}$$

$$r_4 \rightarrow \frac{600069}{23} r_3 + \frac{1}{0.23} r_4$$

Augmented Matrix,

$$\begin{array}{cc} & \begin{array}{ccccc} v_1 & v_2 & v_B & v_L & \end{array} \\ & \begin{bmatrix} -\frac{1}{6K} & \frac{1}{3K} & \frac{1}{0.23} & 0 & 0 \\ 0 & 1 & -1 & -1 & 0.7 \\ 0 & 0 & -\frac{1}{0.23} & \frac{1}{1K} & 0 \\ 0 & 0 & 0 & \frac{900069}{23K} & \frac{790}{23} \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \end{array} \end{array}$$

from equation 4,

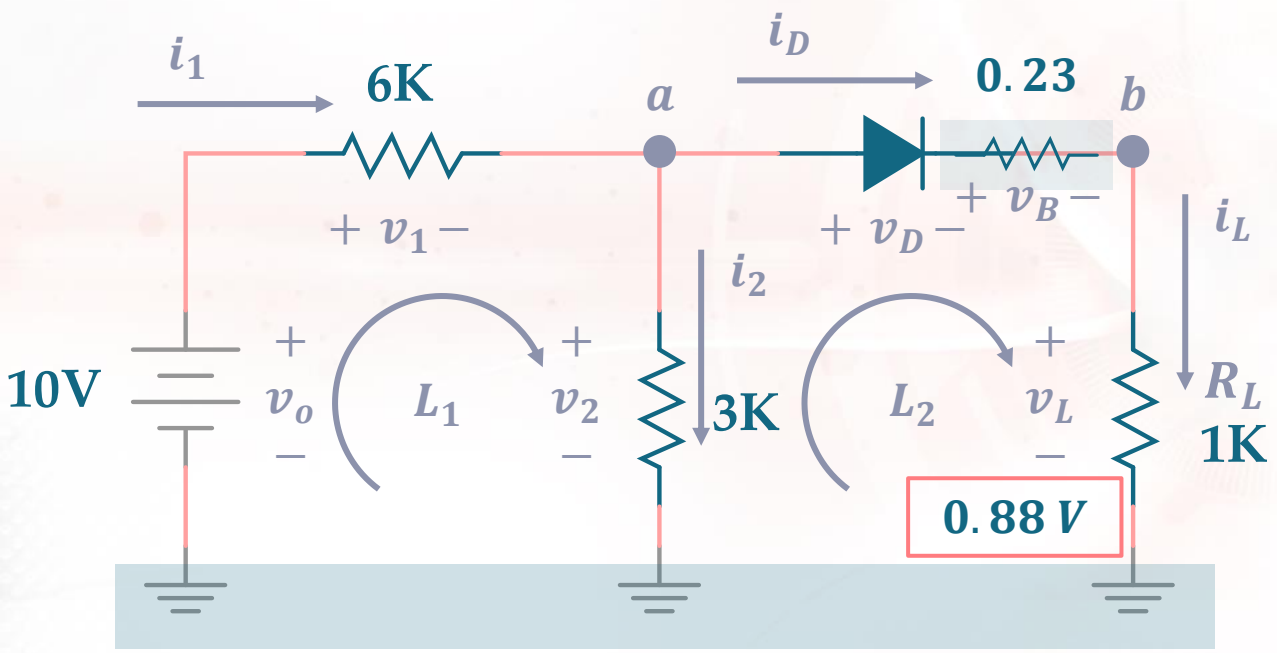
$$\frac{900069}{23K} v_L = \frac{790}{23}$$

$$v_L = \frac{790(23K)}{23(900069)}$$

$$\boxed{v_L = 0.88\ V}$$

ans

The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?



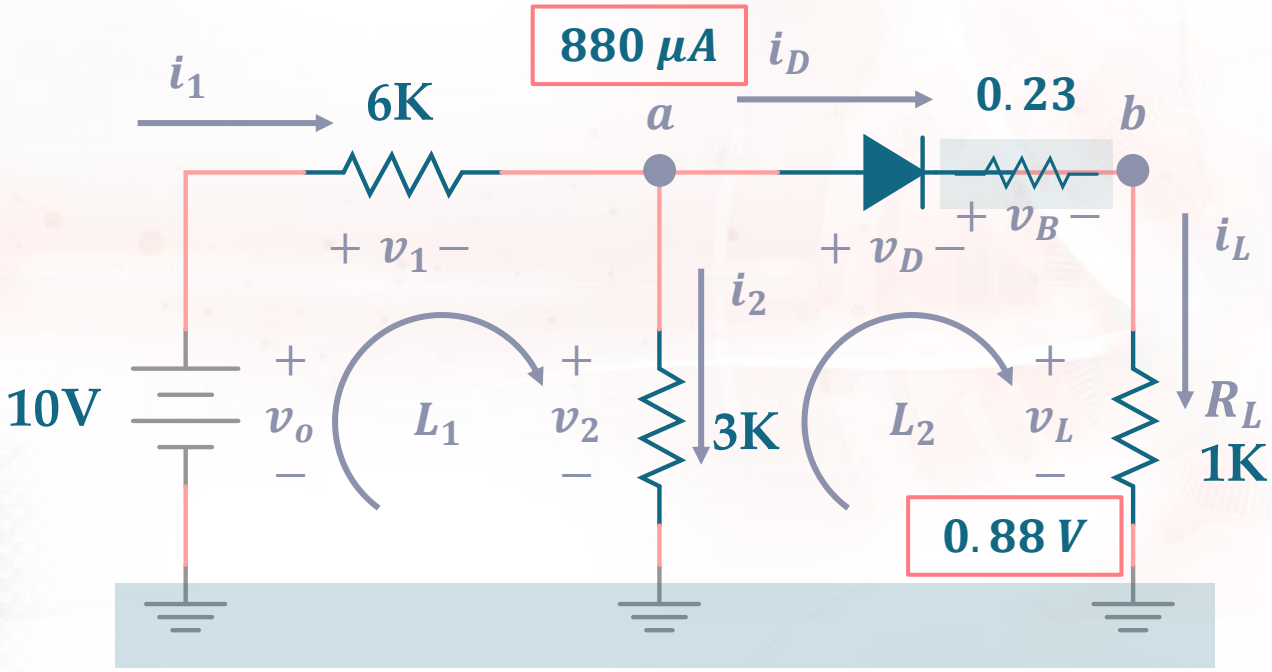
$$i_L = \frac{v_L}{R_L}$$

$$i_L = \frac{0.88}{1K}$$

$$\boxed{i_L = 880\ \mu A}$$

ans

The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?



$$V_D = 0.7 + i_D R_B$$

$$V_D = 0.7 + 880\ uA(0.23)$$

$$V_D = 0.7002$$

$$P_D = i_D v_D$$

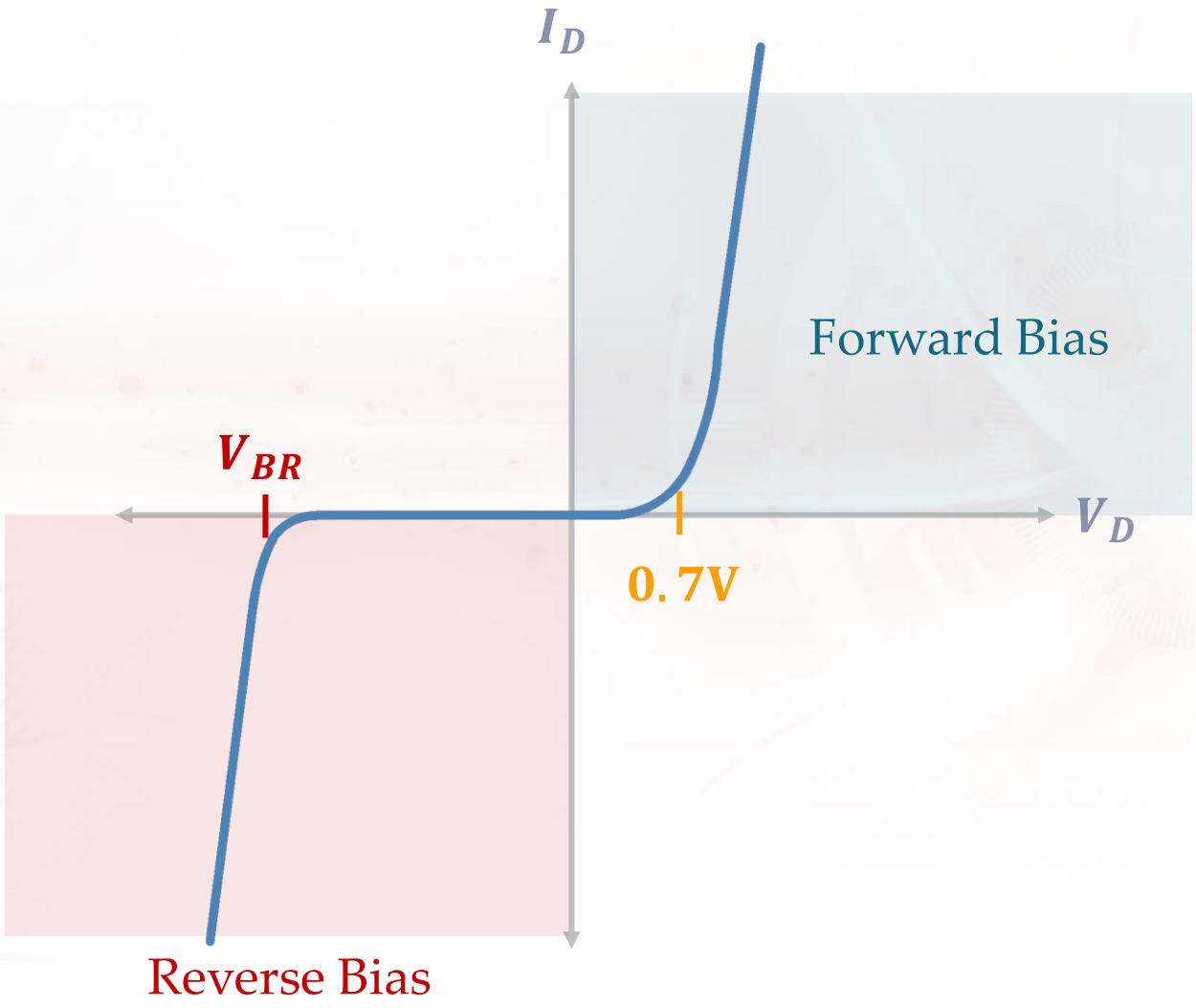
$$P_D = 880\ \mu A(0.7002)$$

$$\boxed{P_D = 616.18\ \mu W}$$

ans

SHOCKLEY'S EQUATION

Diode Characteristic Curve



$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

where,

I_D = forward bias current

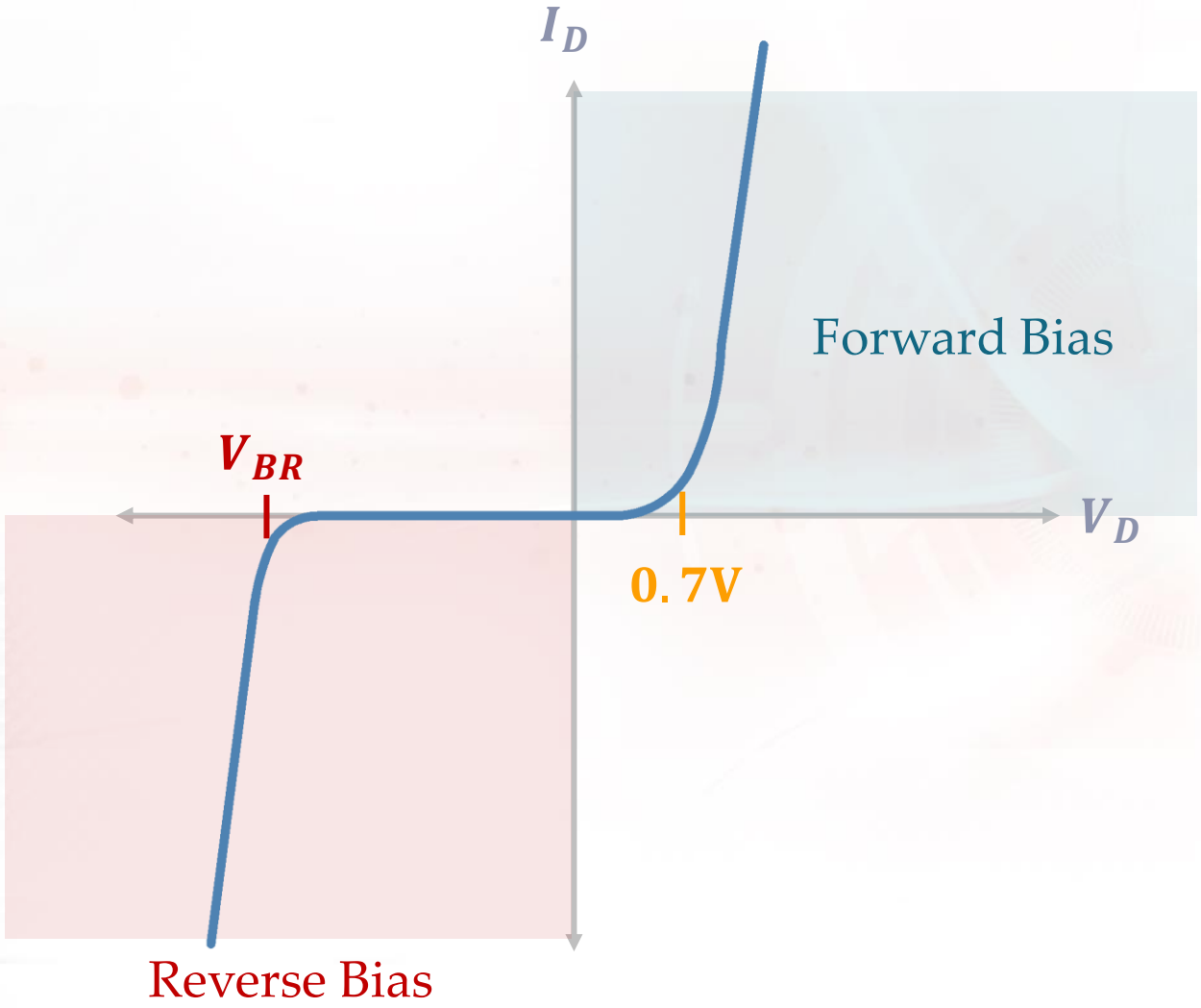
I_S = reverse saturation current

V_D = forward bias diode voltage

n = ideality factor (typ. = 1)

V_T = thermal voltage

Diode Characteristic Curve



$$V_T = \frac{kT}{q}$$

where,

k = Boltzmann's constant
$= 1.38 \times 10^{-23} J/K$

T = absolute temperature
$= \ ^\circ C + 273$

q = electronic charge
$= 1.602 \times 10^{-19} C$

EXERCISE

Determine the **diode current** at **20℃** for silicon diode with $\boldsymbol{I_s = 50nA}$, $\boldsymbol{n = 2}$ and an applied voltage of **0V, 0.5V, 0.7V,** and **1V** .

Determine the **diode current** at **20℃** for silicon diode with $I_s = 50nA$, $n = 2$ and an applied voltage of **0V**, 0.5V, 0.7V, and 1V.

Thermal Voltage,

$$V_T = \frac{kT}{q}$$

$$V_T = \frac{1.38 \times 10^{-23}(20 + 273)}{1.602 \times 10^{-19}}$$

$$V_T = 25.24\ mV$$

Diode current,

$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

$$I_D = 50\ nA\left(e^{\frac{0}{2(25.24\ mV)}} - 1\right)$$

$$\boxed{I_D = 0}$$

ans

Determine the **diode current** at **20°C** for silicon diode with $I_s = 50nA$, $n = 2$ and an applied voltage of 0V, **0.5V**, 0.7V, and 1V.

Thermal Voltage,

$$V_T = \frac{kT}{q}$$

$$V_T = \frac{1.38 \times 10^{-23}(20 + 273)}{1.602 \times 10^{-19}}$$

$$V_T = 25.24\ mV$$

Diode current,

$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

$$I_D = 50\ nA\left(e^{\frac{0.5}{2(25.24\ mV)}} - 1\right)$$

$$\boxed{I_D = 1\ mA}$$

ans

Determine the **diode current** at **20℃** for silicon diode with $I_s = 50nA$, $n = 2$ and an applied voltage of 0V, 0.5V, **0.7V**, and 1V.

Thermal Voltage,

$$V_T = \frac{kT}{q}$$

$$V_T = \frac{1.38 \times 10^{-23}(20 + 273)}{1.602 \times 10^{-19}}$$

$$V_T = 25.24\ mV$$

Diode current,

$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

$$I_D = 50\ nA\left(e^{\frac{0.7}{2(25.24\ mV)}} - 1\right)$$

$$\boxed{I_D = 52.63\ mA}$$

ans

Determine the **diode current** at **20℃** for silicon diode with $I_s = 50nA$, $n = 2$ and an applied voltage of 0V, 0.5V, 0.7V, and **1V**.

Thermal Voltage,

$$V_T = \frac{kT}{q}$$

$$V_T = \frac{1.38 \times 10^{-23}(20 + 273)}{1.602 \times 10^{-19}}$$

$$V_T = 25.24\ mV$$

Diode current,

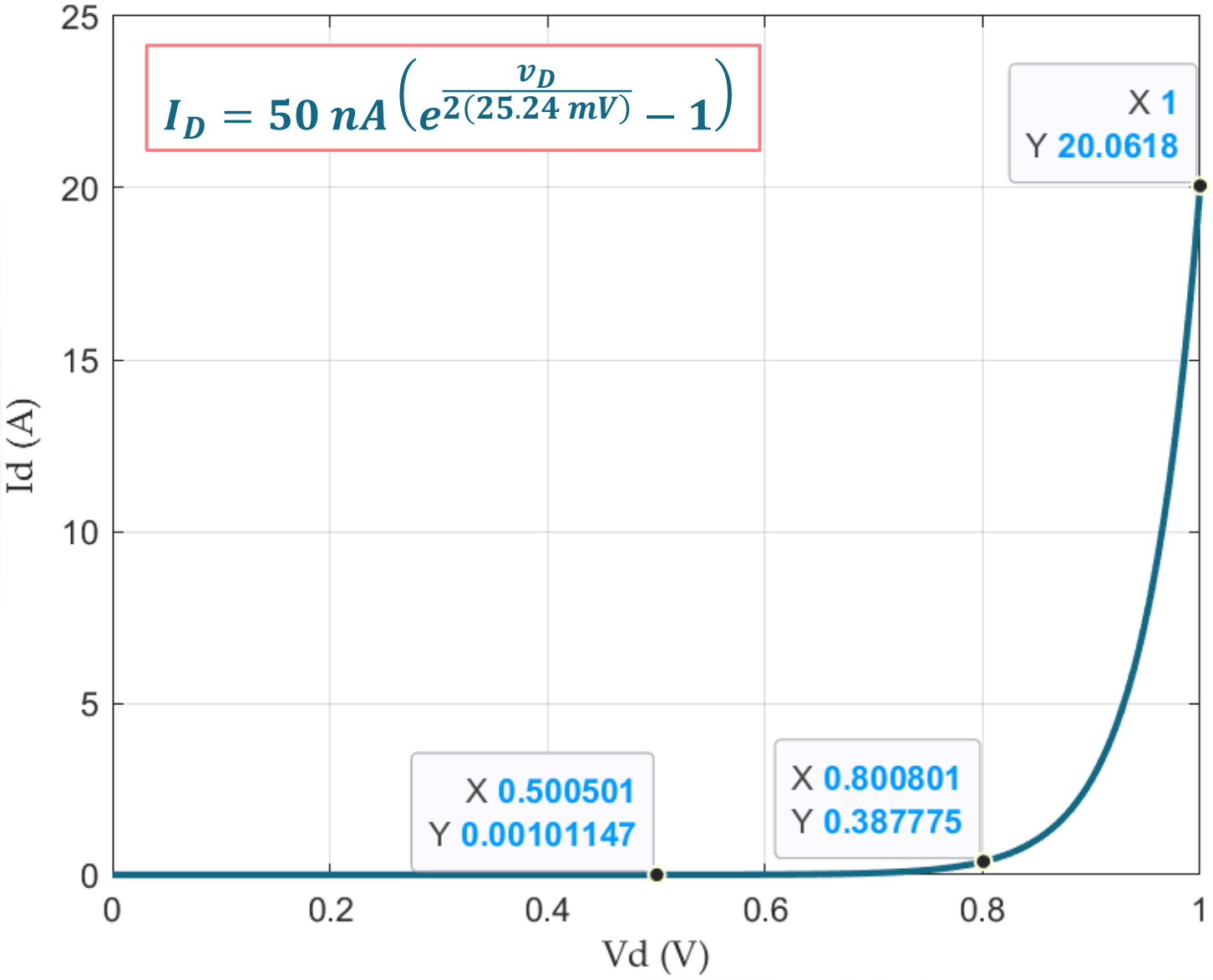
$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

$$I_D = 50\ nA\left(e^{\frac{1}{2(25.24\ mV)}} - 1\right)$$

$$\boxed{I_D = 20.06\ A}$$

ans

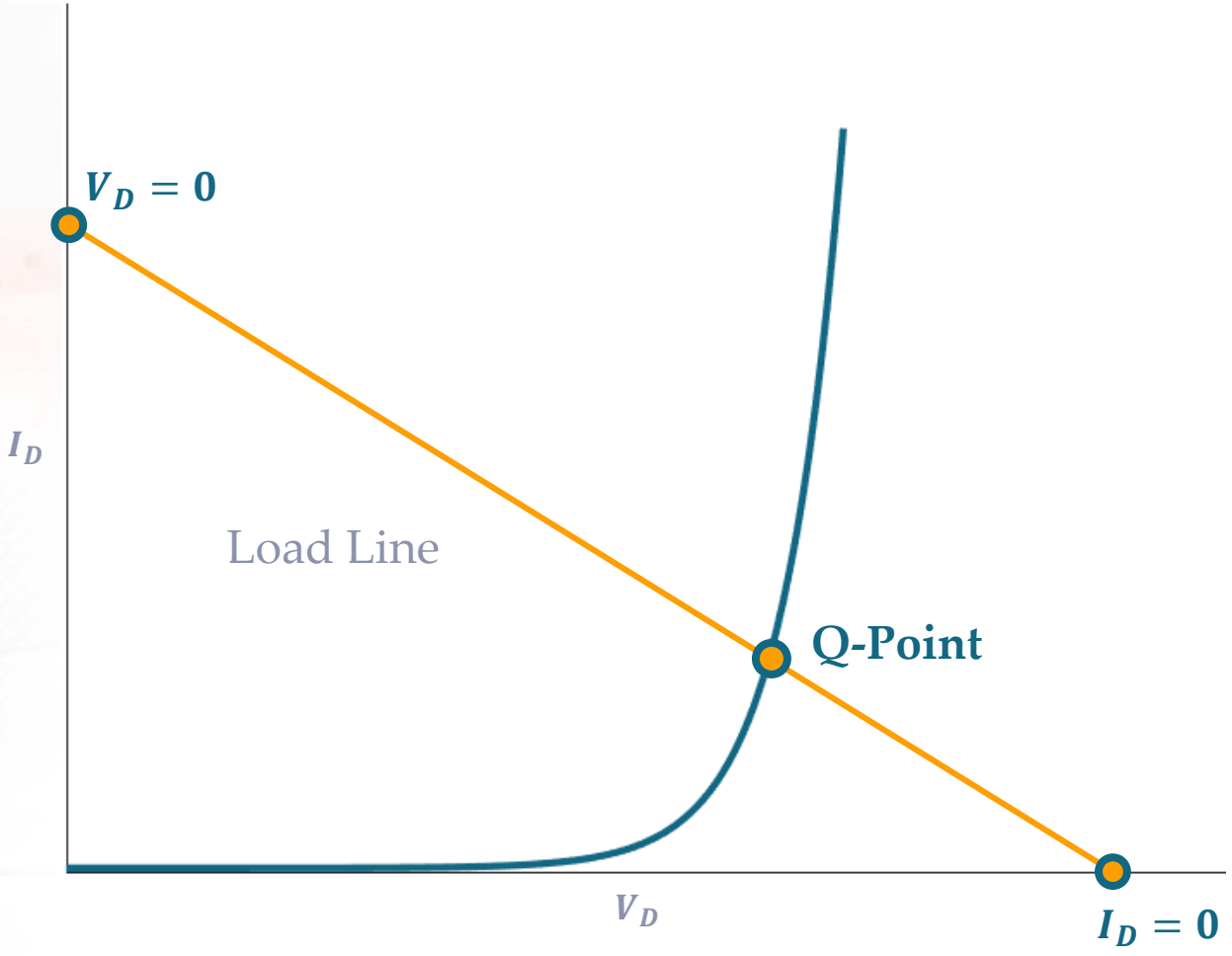
Diode Characteristic Curve



LOAD LINE ANALYSIS

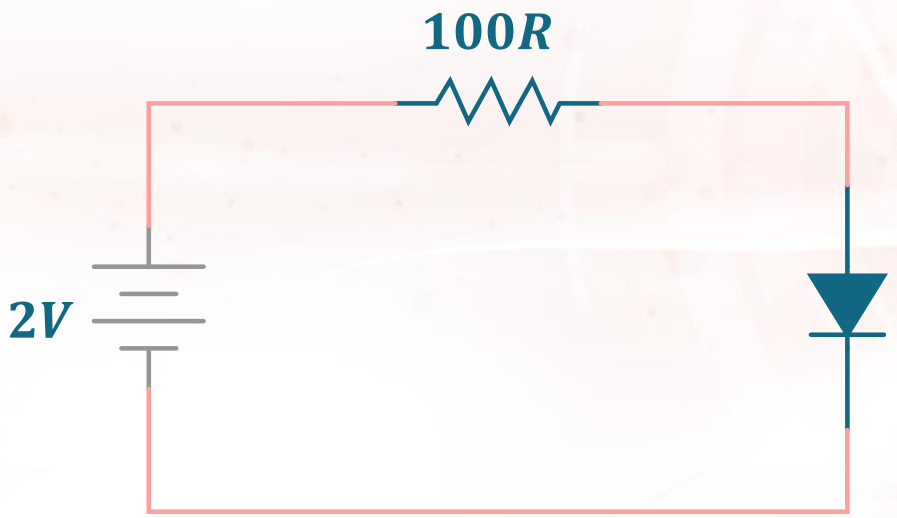
QUIESCENT POINT

Diode Characteristic Curve

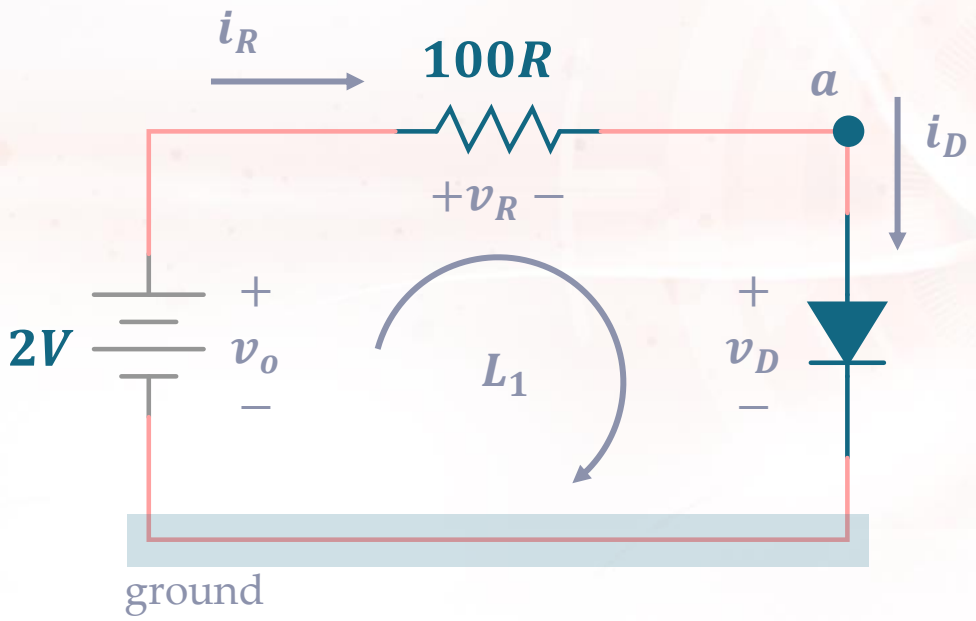


The **intersection** of the load line with the characteristic curve determines the **point of operation** of the system

Determine the point of operation, **Q-Point** of the given circuit.



Determine the point of operation, **Q-Point** of the given circuit.



KCL @ a,

$$-i_R + i_D = 0$$

$$i_R = i_D$$

Let $i_D = 0$,

$$i_R = 0$$

$$\frac{v_R}{R} = 0$$

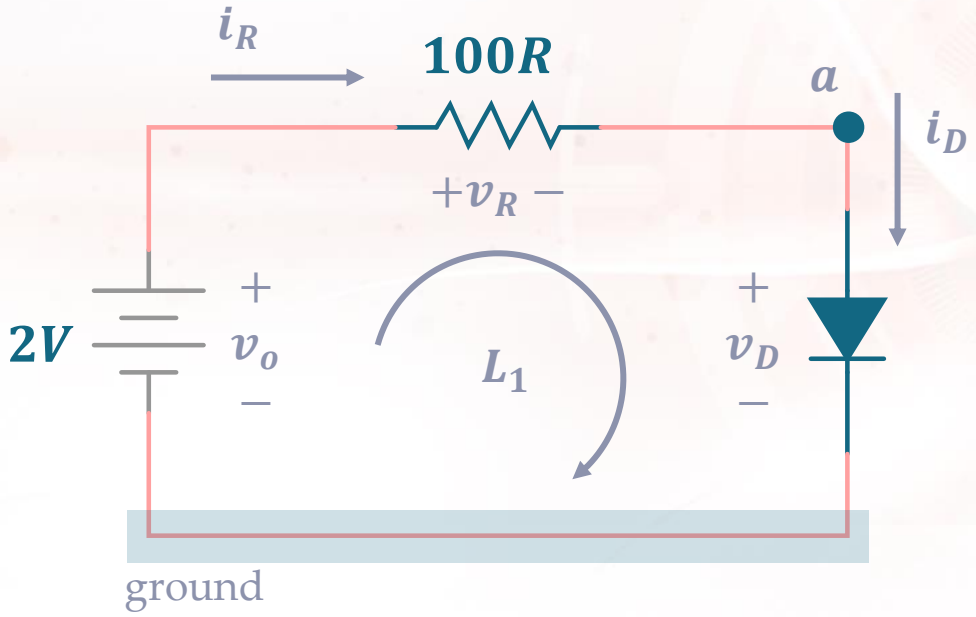
$$\frac{v_o - v_D}{R} = 0$$

$$2 - v_D = 0$$

$$\boxed{v_{D(cutoff)} = 2V}$$

ans

Determine the point of operation, **Q-Point** of the given circuit.



KVL@ L_1,

$$-v_o + v_R + v_D = 0$$

Let $v_D = 0$,

$$-v_o + v_R + 0 = 0$$

$$v_R = v_o$$

$$i_R R = v_o$$

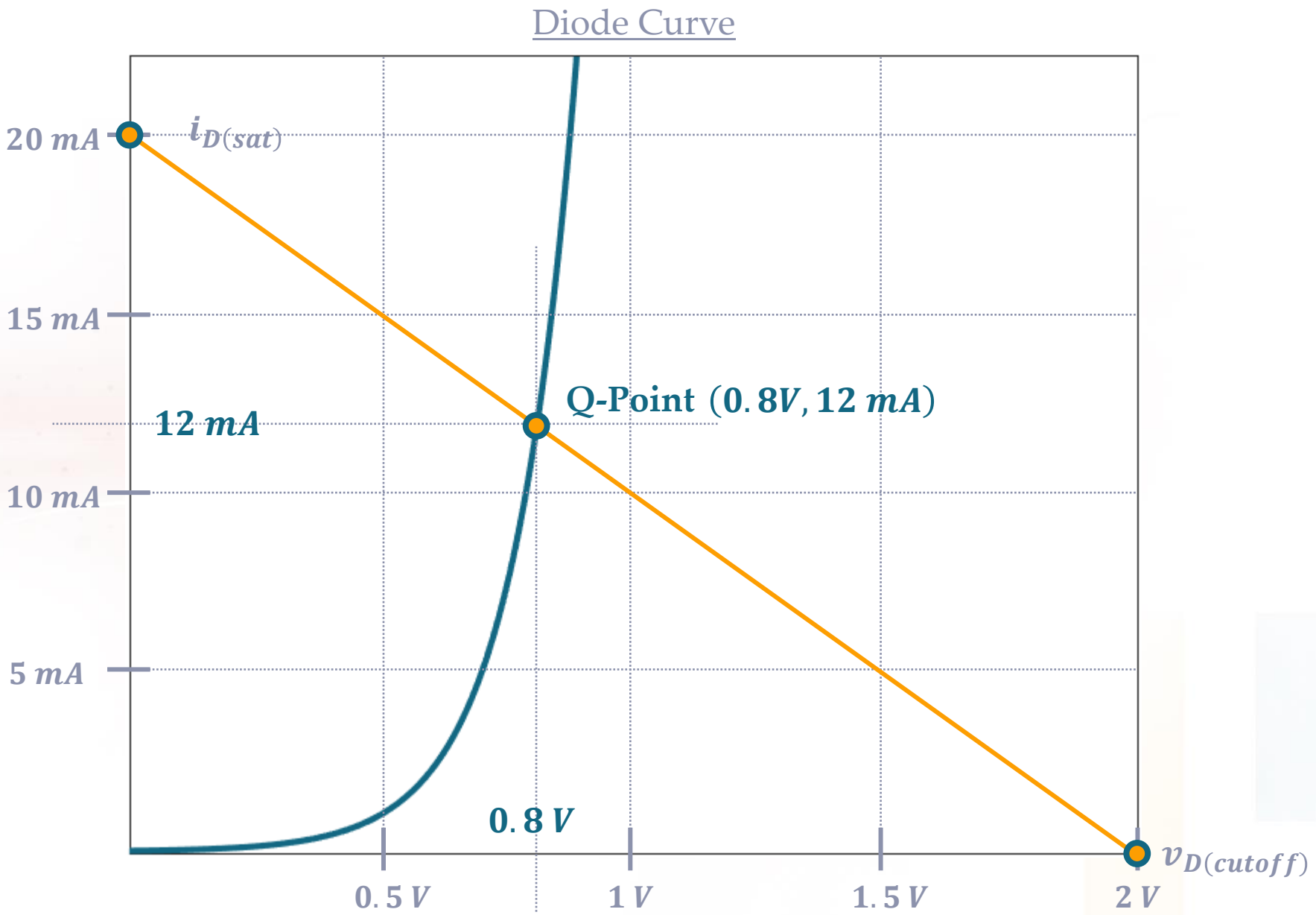
$$i_R = \frac{v_o}{R}$$

$$i_D = \frac{2}{100}$$

$$\boxed{i_{D(sat)} = 20\ mA}$$

ans

LOAD LINE ANALYSIS



READING A DATA SHEET

REVERSE BREADOWN VOLTAGE

MAXIMUM RATINGS

Rating	Symbol	1N4001	1N4002	1N4003	1N4004	1N4005	1N4006	1N4007	Unit
†Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_R	50	100	200	400	600	800	1000	V
†Non–Repetitive Peak Reverse Voltage (halfwave, single phase, 60 Hz)	V_{RSM}	60	120	240	480	720	1000	1200	V
†RMS Reverse Voltage	$V_{R(RMS)}$	35	70	140	280	420	560	700	V
†Average Rectified Forward Current (single phase, resistive load, 60 Hz, T_A = 75°C)	I_O	1.0							A

Destructive level to avoid under all operating conditions

Safety factor of 2

MAXIMUM FORWARD CURRENT

MAXIMUM RATINGS

Rating	Symbol	1N4001	1N4002	1N4003	1N4004	1N4005	1N4006	1N4007	Unit
†Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_R	50	100	200	400	600	800	1000	V
†Non–Repetitive Peak Reverse Voltage (halfwave, single phase, 60 Hz)	V_{RSM}	60	120	240	480	720	1000	1200	V
†RMS Reverse Voltage	$V_{R(RMS)}$	35	70	140	280	420	560	700	V
†Average Rectified Forward Current (single phase, resistive load, 60 Hz, $T_A = 75°C$)	I_O	1.0							A

Level of forward current the diode **burns** out

because of excessive power dissipation

FORWARD VOLTAGE DROP

ELECTRICAL CHARACTERISTICS†

Rating	Symbol	Typ	Max	Unit
Maximum Instantaneous Forward Voltage Drop, (i_F = 1.0 Amp, T_J = 25°C)	v_F	0.93	1.1	V
Maximum Full–Cycle Average Forward Voltage Drop, (I_O = 1.0 Amp, T_L = 75°C, 1 inch leads)	$V_{F(AV)}$	–	0.8	V
Maximum Reverse Current (rated DC voltage) (T_J = 25°C) (T_J = 100°C)	I_R	 0.05 1.0	 10 50	μA
Maximum Full–Cycle Average Reverse Current, (I_O = 1.0 Amp, T_L = 75°C, 1 inch leads)	$I_{R(AV)}$	–	30	μA

The typical voltage across the diode

LABORATORY